TravelGo: A Cloud-Powered Real-Time Travel Booking Platform Using AWS

Project Description:

TravelGo is a full-stack, cloud-based travel booking platform designed to simplify the process of reserving buses, trains, flights, and hotels through a unified interface. Built using Flask as the backend framework, the application is deployed on Amazon EC2 and leverages DynamoDB for efficient storage of user data and bookings. TravelGo allows users to register, log in, search for transportation and accommodation options, and book their travel with ease. Once a booking is confirmed or cancelled, users receive real-time email notifications powered by AWS Simple Notification Service (SNS), keeping them informed throughout their journey.

The platform's user-friendly interface supports dynamic seat selection for buses, hotel filtering based on preferences such as luxury or budget, and provides booking summaries along with centralized cancellation management. By combining cloud scalability, responsive design, and secure session handling, TravelGo delivers a seamless and real-time travel planning experience for users.

Scenario 1: Hassle-Free Multi-Mode Travel Booking Experience

TravelGo offers users a unified platform to search and book buses, trains, flights, and hotels all in one place. For instance, a user planning a trip from Hyderabad to Bangalore can log in, select their preferred mode of transport, choose from available options, and proceed to booking. Flask manages the backend operations such as retrieving travel listings and processing user input in real-time.
Hosted on AWS EC2, the platform remains responsive even during high-traffic hours like weekends or holiday seasons, allowing multiple users to browse and book without delay.

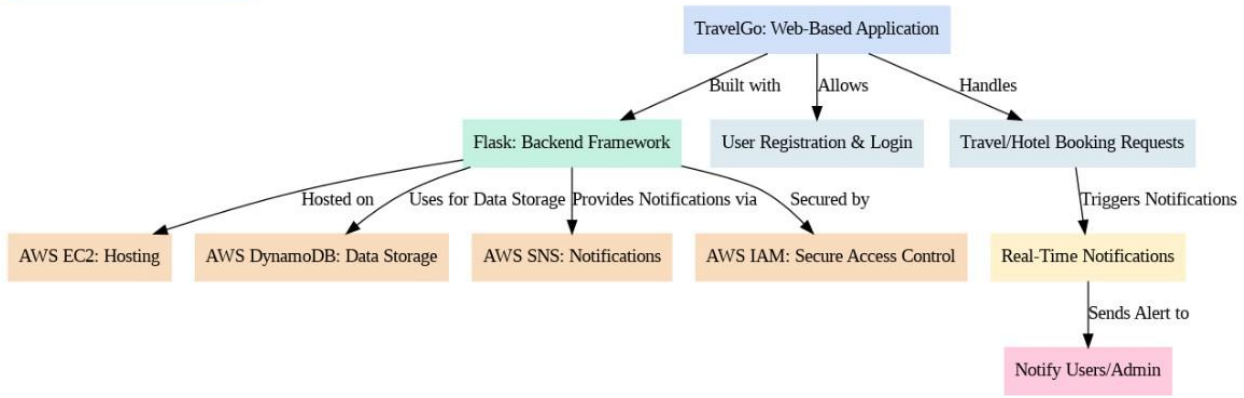
Scenario 2: Real-Time Booking Confirmation with AWS SNS

Once a booking is made—whether it's a train ticket or a hotel stay—TravelGo uses AWS SNS to instantly notify the user. For example, after a student books a hotel in Chennai, SNS sends a real-time email notification confirming the booking with all the relevant details. This notification is triggered from the Flask backend after the booking is successfully recorded in DynamoDB. Additionally, SNS can alert admin or service providers, ensuring transparency and real-time updates on every transaction.

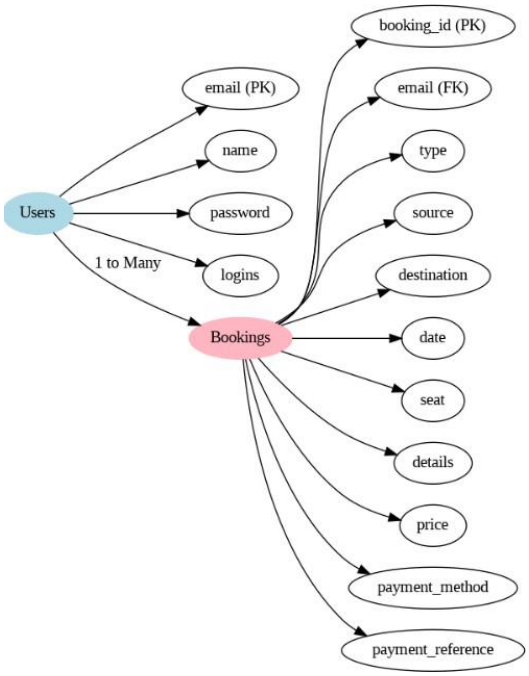
Scenario 3: Dynamic Dashboard with Personal Travel History

TravelGo features a dynamic user dashboard that displays all past and upcoming bookings for the logged-in user. For example, a user who has booked a flight and a hotel can view these bookings categorized by type, along with dates, price, and cancellation options. Flask fetches this data from AWS DynamoDB, which persistently stores all user bookings. The dashboard UI, powered by responsive HTML/CSS and Flask templates, ensures users can review or manage bookings anytime, from any device, with real-time updates and quick cancellation workflows supported.

AWS ARCHITECTURE



Entity Relationship (ER)Diagram:



Pre-requisites:

1. **AWS Account Setup**: AWS Account Setup
2. **Understanding IAM**: IAM Overview
3. **Amazon EC2 Basics**: EC2 Tutorial
4. **DynamoDB Basics**: DynamoDB Introduction
5. **SNS Overview**: SNS Documentation
6. **Git Version Control**: Git Documentation

Project WorkFlow:

1. AWS Account Setup and Login

Activity 1.1: Set up an AWS account if not already done.

Activity 1.2: Log in to the AWS Management Console

2. DynamoDB Database Creation and Setup

Activity 2.1: Create a DynamoDB Table.

Activity 2.2: Configure Attributes for travel-Users and Bookings Tables.

3. SNS Notification Setup

Activity 3.1: Create an SNS Topic named BookingConfirmation for sending notifications.

Activity 3.2: Subscribe email to the topic to receive booking and cancellation alerts.

4. Backend Development and Application Setup

Activity 4.1:Develop the Backend Using Flask.

Activity 4.2: Integrate AWS Services Using boto3.

5. IAM Role Setup

Activity 5.1: Create IAM Role

Activity 5.2: Attach Policies

6. EC2 Instance Setup

Activity 6.1: Launch an EC2 instance to host the Flask application.

Activity 6.2: Configure security groups for HTTP, and SSH access.

7. Deployment on EC2

Activity 7.1:Upload Flask Files

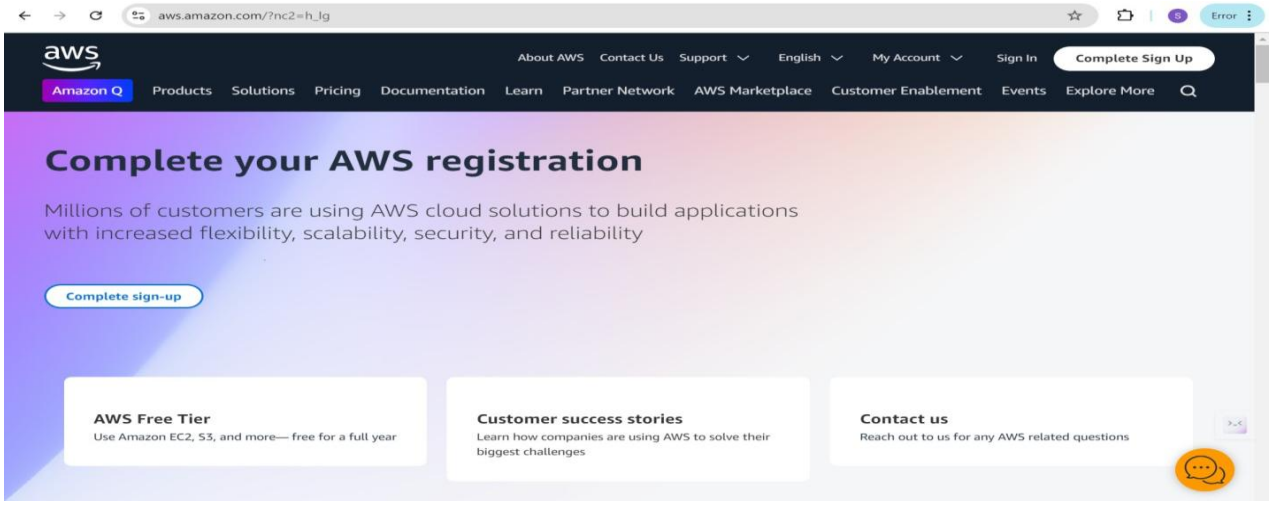
Activity 7.2: Run the Flask App

8. Testing and Deployment

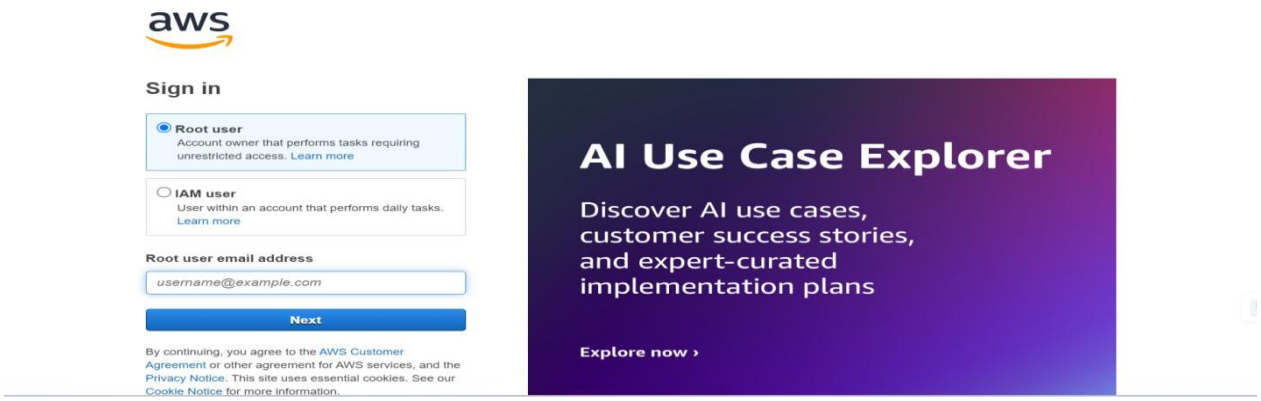
Activity 8.1: Conduct functional testing to verify user registration, login, book requests, and notifications.

Milestone 1: AWS Account Setup and Login

- **Activity 1.1: Set up an AWS account if not already done.**
  - Sign up for an AWS account and configure billing settings.



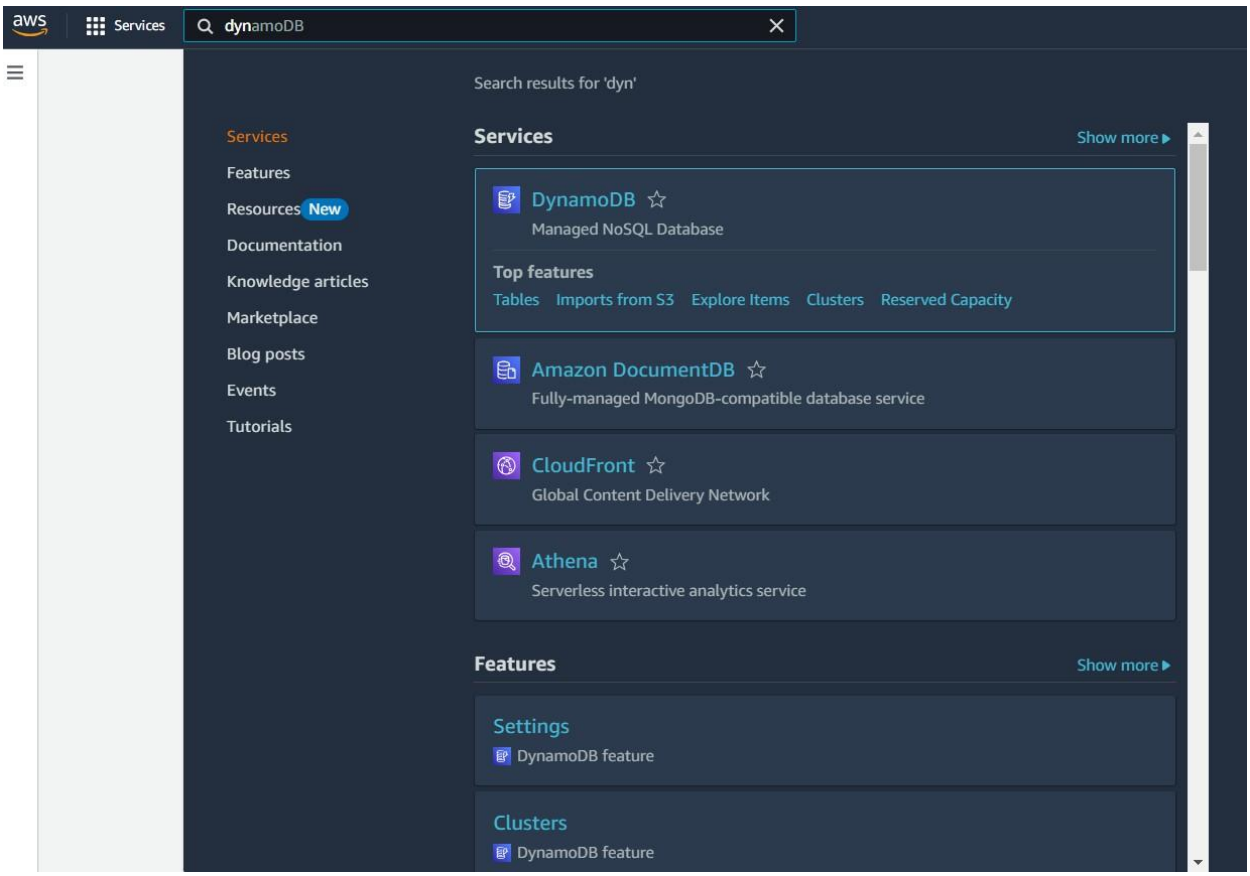
- **Activity 1.2: Log in to the AWS Management Console**
  - After setting up your account, log in to the AWS Management Console.



Milestone 2: DynamoDB Database Creation and Setup

- **Activity 2.1:Navigate to the DynamoDB**

- In the AWS Console, navigate to DynamoDB and click on create tables.



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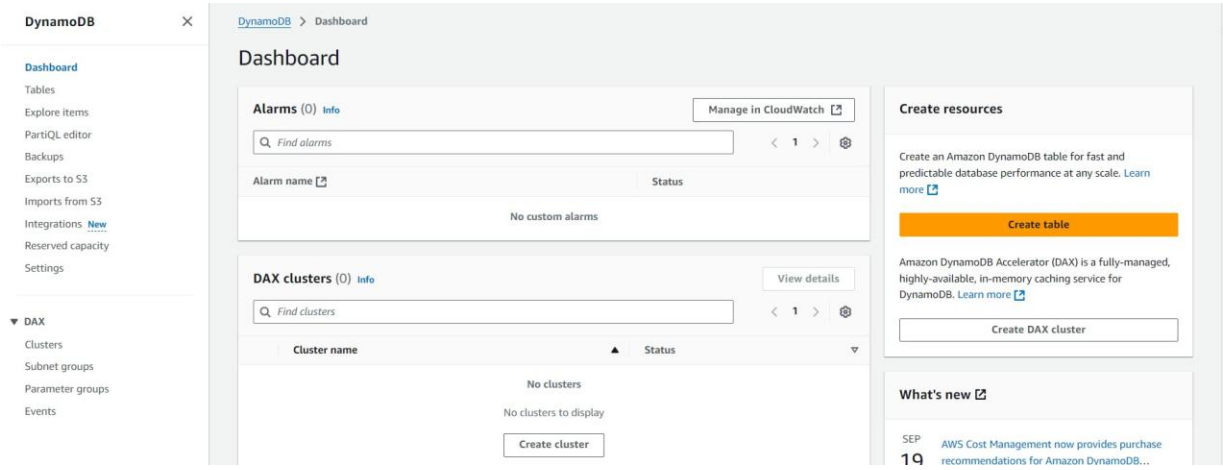
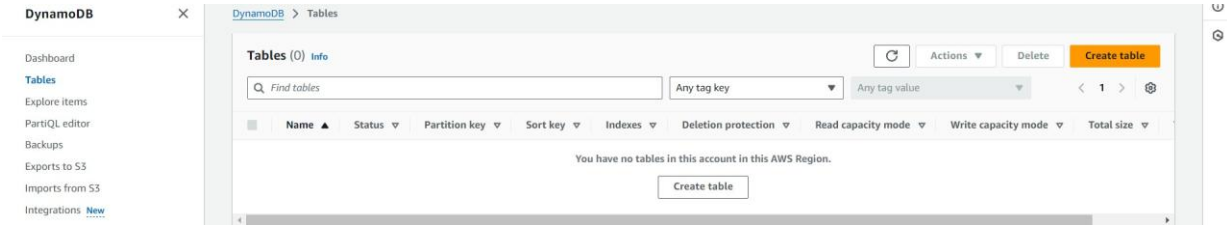
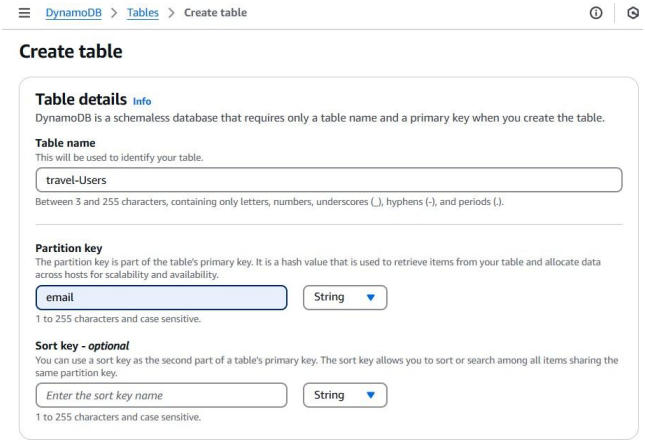




- **Activity 2.2: Create DynamoDB tables for storing registration details and booking records.**
    - Create a travel-Users table for storing registered user details with partition key "Email" with type String and click on create tables.



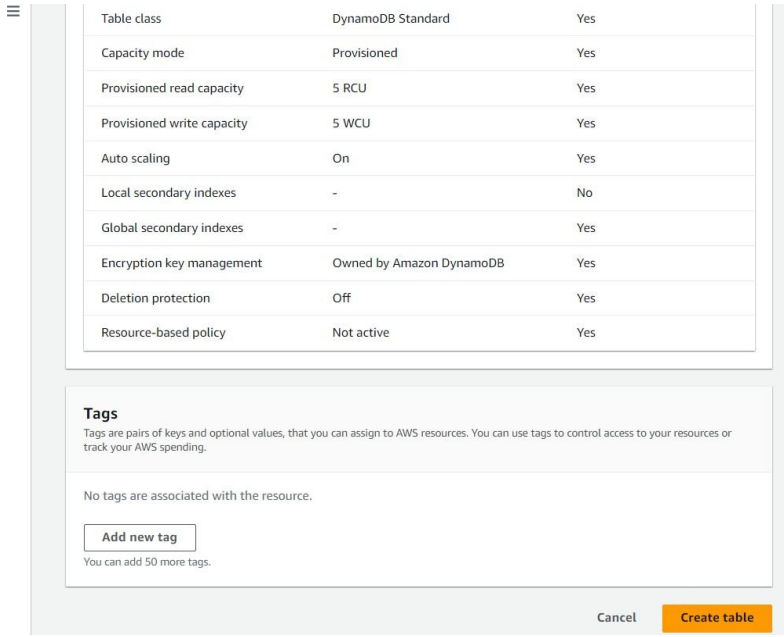
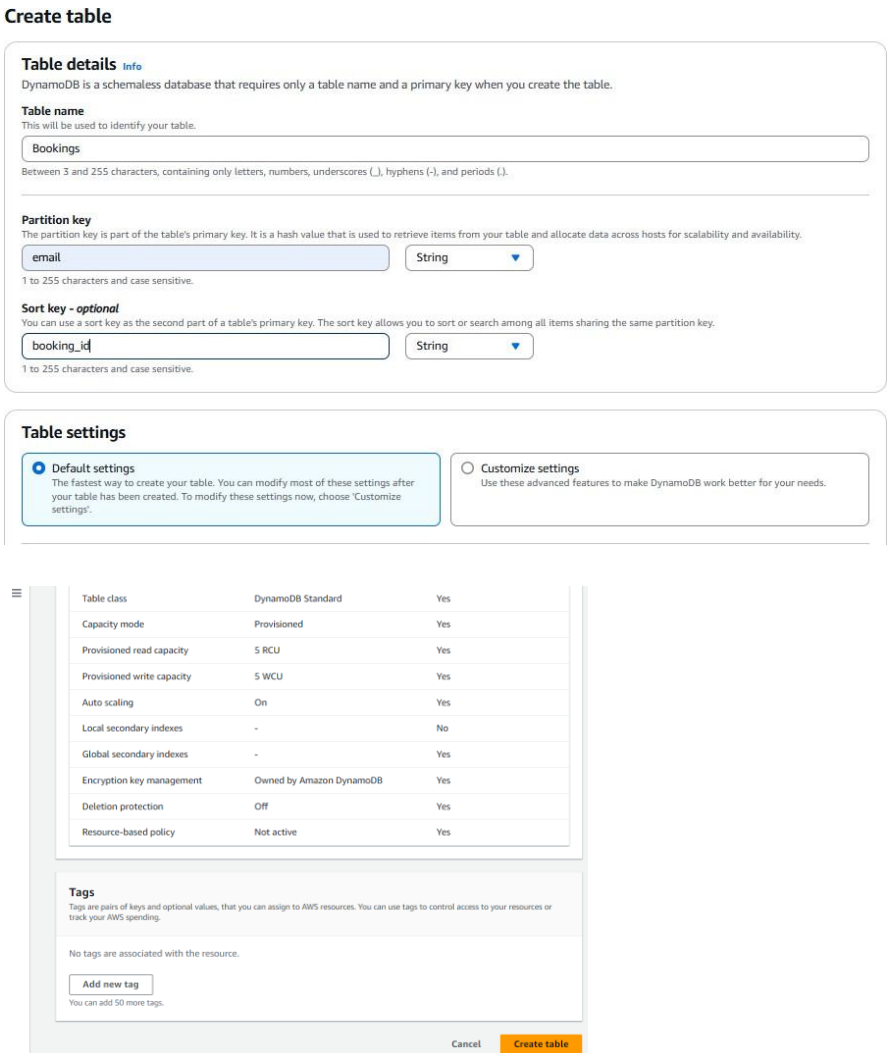
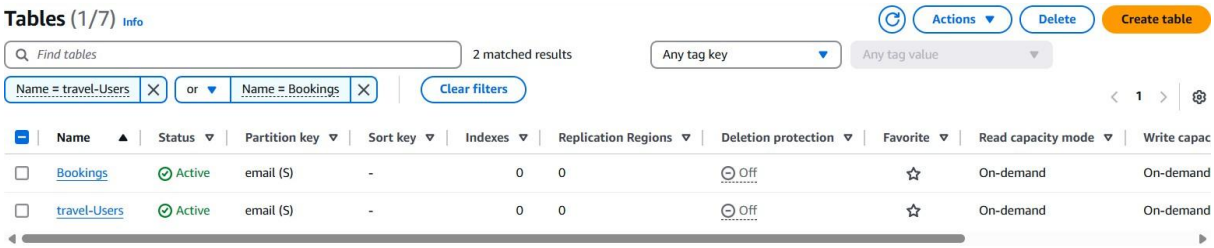


Table class	DynamoDB Standard	Yes
Capacity mode	Provisioned	Yes
Provisioned read capacity	5 RCU	Yes
Provisioned write capacity	5 WCU	Yes
Auto scaling	On	Yes
Local secondary indexes	-	No
Global secondary indexes	-	Yes
Encryption key management	Owned by Amazon DynamoDB	Yes
Deletion protection	Off	Yes
Resource-based policy	Not active	Yes

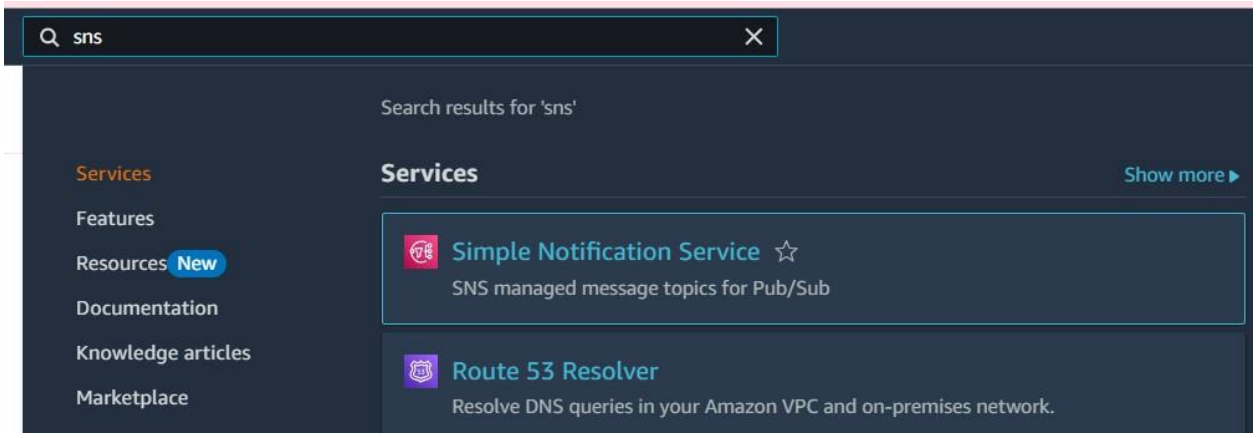
- Follow the same steps to create a `Bookings` for storing booking records with `email` as the partition key and `booking_id` as the sort key.

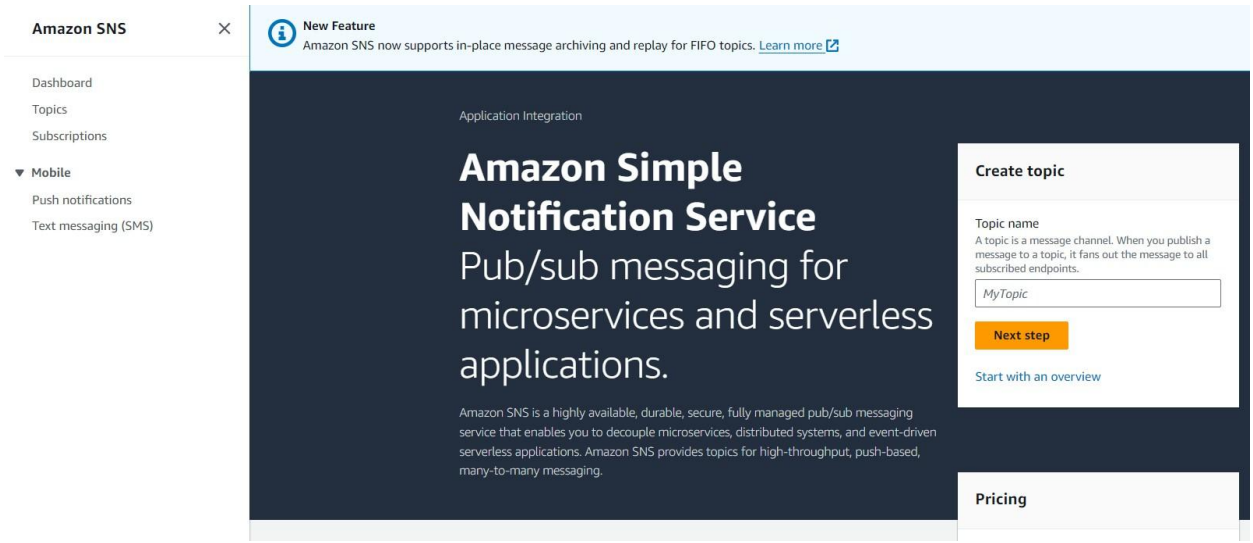




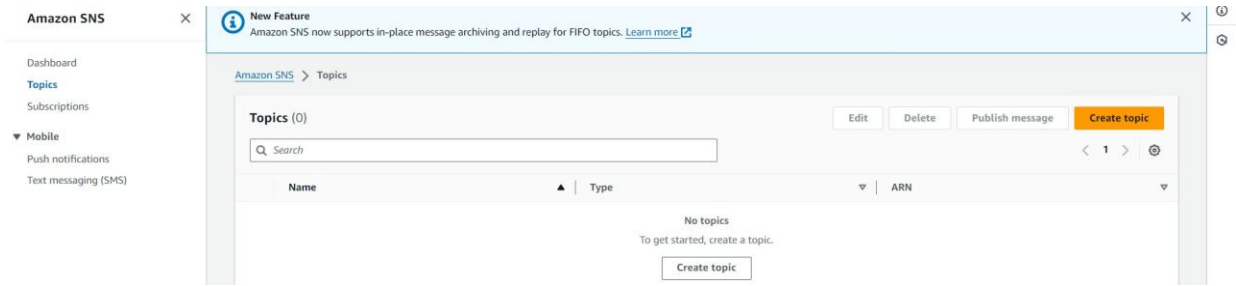
Milestone 3: SNS Notification Setup

- **Activity 3.1: Create an SNS Topic named BookingConfirmation for sending notifications.**
    - In the AWS Console, search for SNS and navigate to the SNS Dashboard.

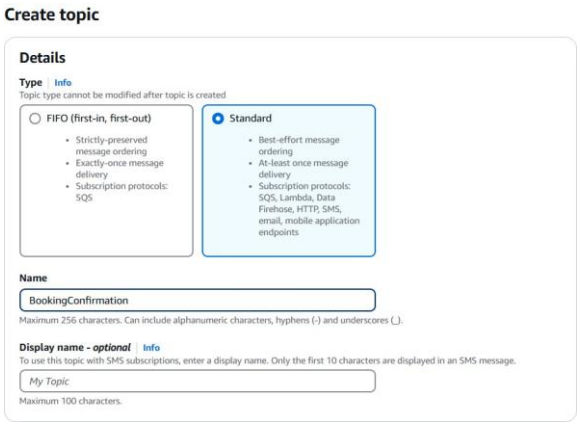


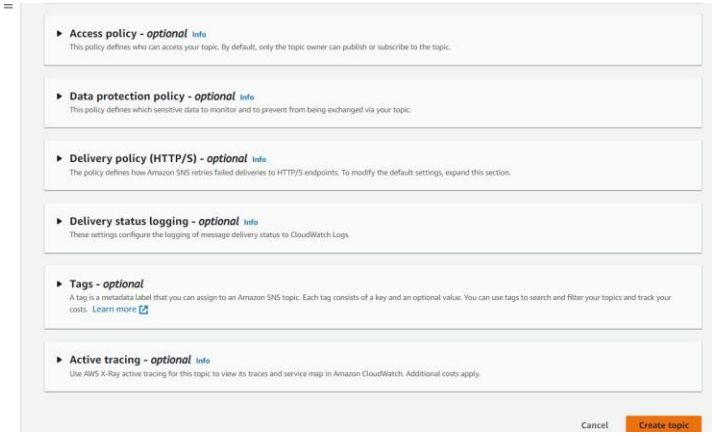


- Click on **Create Topic** and choose a name for the topic.

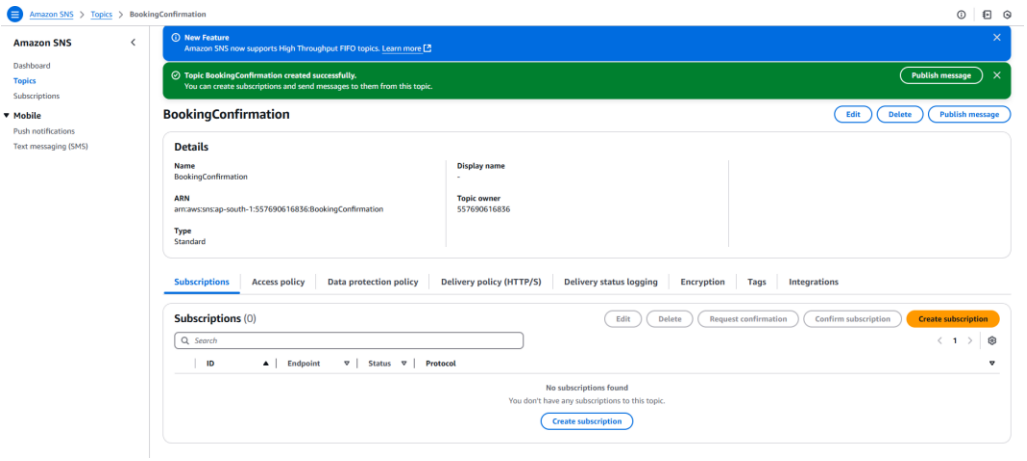


- Choose Standard type for general notification use cases and Click on Create Topic.

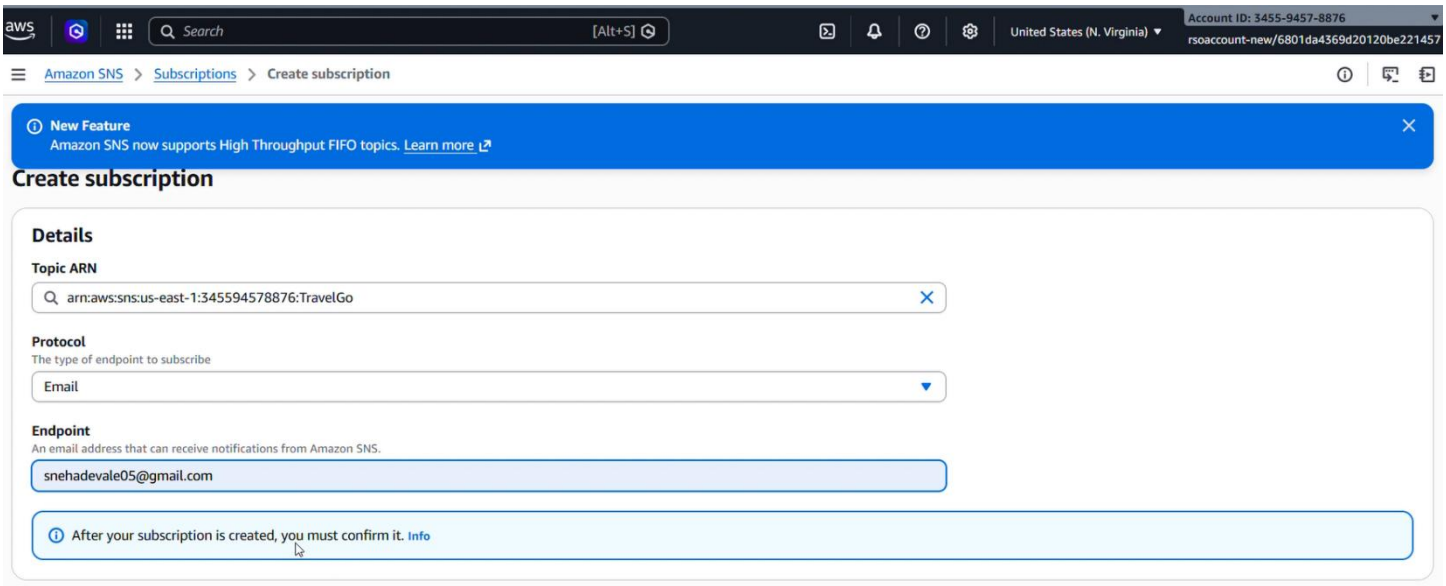




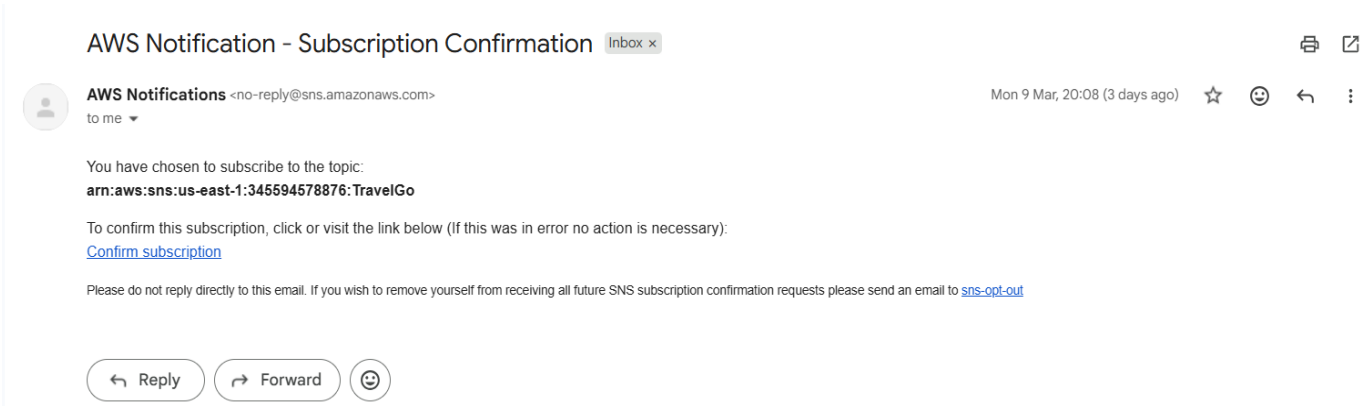
- Configure the SNS topic and note down the **Topic ARN**.

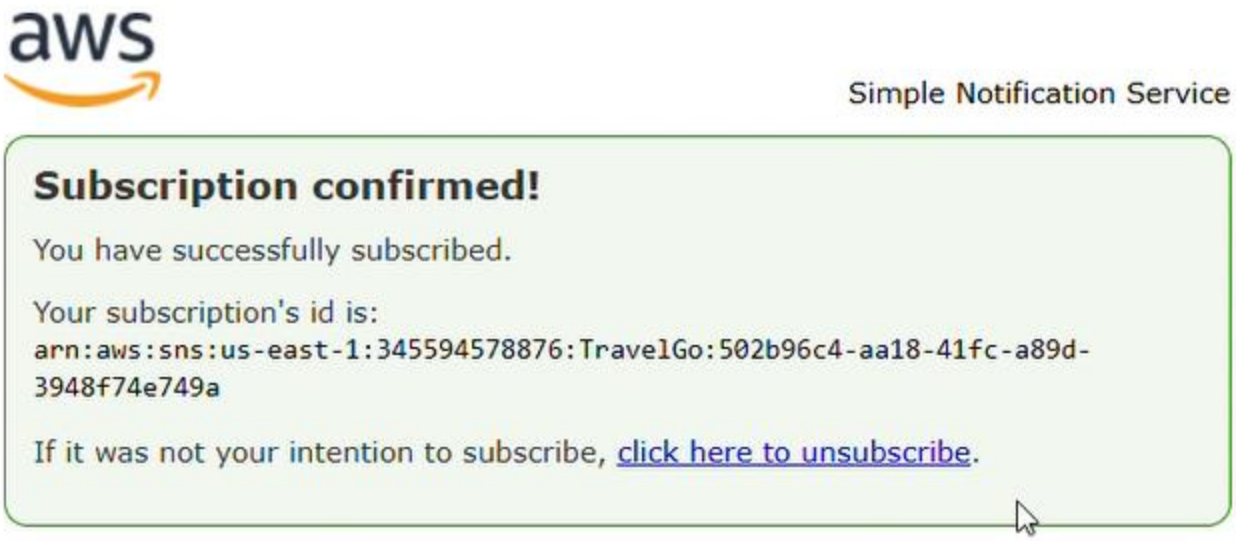


- **Activity 3.2: Subscribe email to the topic to receive booking and cancellation alerts.**
    - Subscribe users (or admin staff) to this topic via Email. When a book request is made, notifications will be sent to the subscribed emails**.**
    - After subscription request for the mail confirmation



- Navigate to the subscribed Email account and Click on the confirm subscription in the AWS Notification- Subscription Confirmation mail.

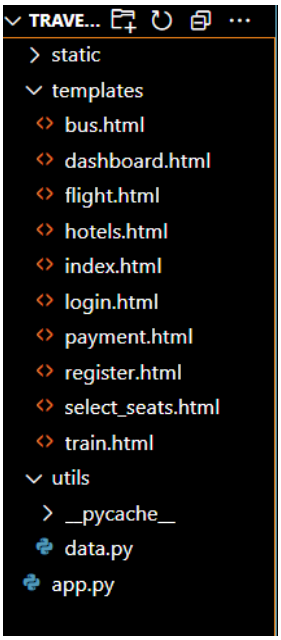




- Successfully done with the SNS mail subscription and setup, now store the ARN link.

Milestone 4:Backend Development and Application Setup

- **Activity 4.1: Develop the backend using Flask**
  - File Explorer Structure:



Description: Create a `templates` folder containing HTML files for each major feature—bus, train, flight, hotels, seat selection, registration, login, payment, and dashboard—and manage backend logic in `app.py` and `utils/data.py`.

Description of the code :

- **Flask App Initialization**

Import Libraries:

```
from flask import Flask, render_template, request, redirect, session
import uuid
import boto3
import os
from decimal import Decimal
from boto3.dynamodb.conditions import Attr
from utils.data import transport_data, hotel_data
```

Description: Import necessary Flask modules for web routing, Boto3 for AWS DynamoDB access, UUID for unique IDs, and custom transport and hotel data from `utils.data`

Initialize flask app:

```
app = Flask(__name__)
```

Description: Initialize the Flask application by creating an instance of the `Flask` class with the current module name.

DynamoDB Connection:

```
# AWS Setup
dynamodb = boto3.resource('dynamodb', region_name='ap-south-1')


# Tables
users_table = dynamodb.Table('travel-Users')
bookings_table = dynamodb.Table('Bookings')
```

Description: Use **boto3** to connect to **DynamoDB** for handling travel-users, booking database operations and also mention region_name where Dynamodb tables are created.

SNS Connection:

```
#sns setup

sns = boto3.client('sns', region_name='ap-south-1')
```

Description: Configure **SNS** to send notifications when a booking request is submitted.

- **Routes for Web Pages**

Register User:

```
@app.route('/')
def home():
    return render_template('index.html', logged_in='user' in session)
```

```
@app.route('/register', methods=['GET', 'POST'])
def register():
    if request.method == 'POST':
        email = request.form['email']
        response = users_table.get_item(Key={'email': email})
        if 'Item' in response:
            return render_template('register.html', message="User already
        users_table.put_item(Item={
            'email': email,
            'name': request.form['name'],
            'password': request.form['password'],
            'logins': 0
        })
        return redirect('/login')
    return render_template('register.html')
```

Description: Collecting registration data, and stores user details in the database.

- **login Route (GET/POST)**:

```
@app.route('/login', methods=['GET', 'POST'])
def login():
    if request.method == 'POST':
        email = request.form['email']
        password = request.form['password']
        response = users_table.get_item(Key={'email': email})
        user = response.get('Item')
        if user and user['password'] == password:
            session['user'] = email
            users_table.update_item(
                Key={'email': email},
                UpdateExpression="ADD logins :inc",
                ExpressionAttributeValues={':inc': Decimal(1)}
            )
            return '''
                <script>
                    localStorage.setItem("loggedIn", "true");
                    window.location.href = "/dashboard";
                </script>
            '''
        return render_template('login.html', message="Invalid credentials.
    return render_template('login.html')
```

Description: Verifies user credentials, increments login count, and redirects to the dashboard on success.

Dashboard route:

```
@app.route('/dashboard')
def dashboard():
    if 'user' not in session:
        return redirect('/login')
    email = session['user']
    user = users_table.get_item(Key={'email': email}).get('Item')
    response = bookings_table.scan(FilterExpression=Attr('user_email').eq(email))
    return render_template('dashboard.html', name=user['name'], bookings=response['Items'])
```

Description: Define the `/dashboard` route to display the user's booking data from DynamoDB after verifying the session and fetching user details using their email.

Transport Routes:

```python
# ---------- Transport Pages ----------
@app.route('/bus', methods=['GET', 'POST'])
def bus():
    if request.method == 'POST':
        s, d, dt = request.form['source'], request.form['destination'], request.form['date']
        buses = [b for b in transport_data['bus'] if b['source'] == s and b['destination'] == d]
        return render_template('bus.html', buses=buses, source=s, destination=d, date=dt)
    return render_template('bus.html', buses=None)

@app.route('/train', methods=['GET', 'POST'])
def train():
    if request.method == 'POST':
        s, d, dt = request.form['source'], request.form['destination'], request.form['date']
        trains = [t for t in transport_data['train'] if t['source'] == s and t['destination'] == d]
        return render_template('train.html', trains=trains, source=s, destination=d, date=dt)
    return render_template('train.html', trains=None)

@app.route('/flight', methods=['GET', 'POST'])
def flight():
    if request.method == 'POST':
        s, d, dt = request.form['source'], request.form['destination'], request.form['date']
        flights = [f for f in transport_data['flight'] if f['source'] == s and f['destination'] == d]
        return render_template('flight.html', flights=flights, source=s, destination=d, date=dt)
    return render_template('flight.html', flights=None)

@app.route('/hotels', methods=['GET', 'POST'])
def hotels():
    if request.method == 'POST':
        city = request.form['city']
        hotels = [h for h in hotel_data if h['location'] == city]
        return render_template('hotels.html', hotels=hotels, city=city)
    return render_template('hotels.html', hotels=None)
```

```python
@app.route('/select_seats')
def select_seats():
    return render_template("select_seats.html")
```

Description: Define separate route handlers for bus, train, flight, and hotel search functionalities using POST requests to filter results from local datasets based on user input like source, destination, date, or city and also add seat selecting routes for bus page.

Booking routes:

```python
# ---------- Booking ----------
@app.route('/book', methods=['POST'])
def book():
    if 'user' not in session:
        return redirect('/login')

    session['pending_booking'] = {
        "booking_id": str(uuid.uuid4())[:8],
        "type": request.form['type'],
        "source": request.form['source'],
        "destination": request.form['destination'],
        "date": request.form['date'],
        "seat": request.form.get('seat', 'N/A'),
        "details": request.form['details'],
        "price": Decimal(request.form['price']),
        "user_email": session['user'],
        "email": session['user']
    }
    return render_template("payment.html", booking=session['pending_booking'])
```

Description: Saves booking details in session and redirects to the payment page if the user is logged in.

Payment Route:

```python
@app.route('/payment', methods=['POST'])
def payment():
    if 'user' not in session or 'pending_booking' not in session:
        return redirect('/login')

    booking = session.pop('pending_booking')
    booking['payment_method'] = request.form['method']
    booking['payment_reference'] = request.form['reference']

    bookings_table.put_item(Item=booking)

    # SNS Booking Notification
    try:
        sns.publish(
            TopicArn=os.getenv('SNS_TOPIC_ARN'),
            Message=f"✅ Booking Confirmed!\n{booking['details']} on {booking['date']} for ₹{booking['price']}",
            Subject="TravelGo Booking Confirmation"
        )
    except Exception as e:
        print("SNS Error (booking):", e)

    return redirect('/dashboard')
```

Description: Processes payment details, stores the booking in the database, and sends an SNS confirmation notification before redirecting to the dashboard.

Remove Booking route:

```
@app.route('/remove_booking', methods=['POST'])
def remove_booking():
    if 'user' not in session:
        return redirect('/login')
    email = session['user']
    booking_id = request.form.get('booking_id')

    try:
        bookings_table.delete_item(Key={'email': email, 'booking_id': booking_id})
        # SNS Cancellation Notification
        sns.publish(
            TopicArn=os.getenv('SNS_TOPIC_ARN'),
            Message=f"❌ Booking Canceled\nBooking ID: {booking_id}",
            Subject="TravelGo Cancellation"
        )
    except Exception as e:
        print("Delete Error:", e)

    return redirect('/dashboard')
```

Description: Deletes a booking from the database and sends a cancellation notification via SNS, then redirects to the dashboard.

Logout route:

```
@app.route('/logout')
def logout():
    session.clear()
    return '''
        <script>
            localStorage.setItem("loggedIn", "false");
            window.location.href = "/";
        </script>
    '''
```

Description: Implement the logout route to clear session data and reset `loggedIn` in localStorage before redirecting the user to the homepage.

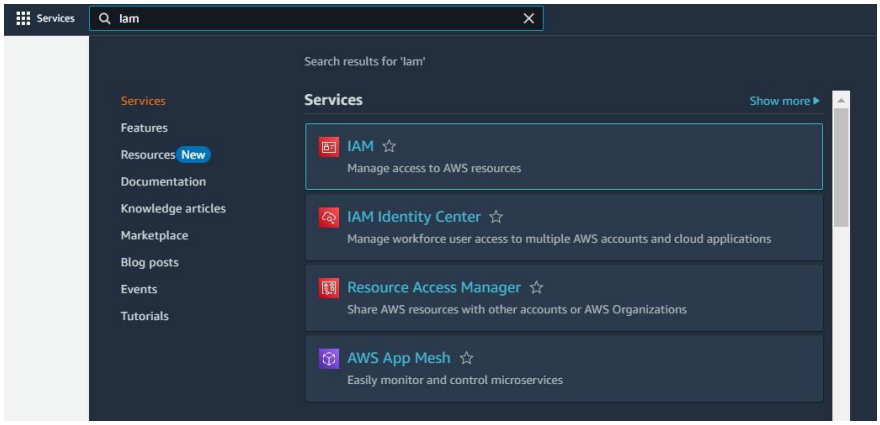
Deployment Code:

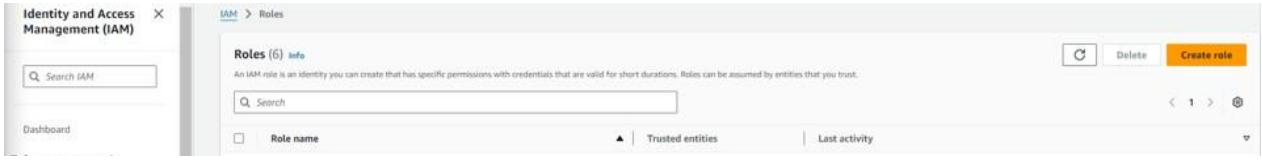
```
if __name__ == '__main__':
    # Running on 0.0.0.0 for EC2 access, but debug is OFF for safety
    app.run(host='0.0.0.0', port=5000, debug=False)
```

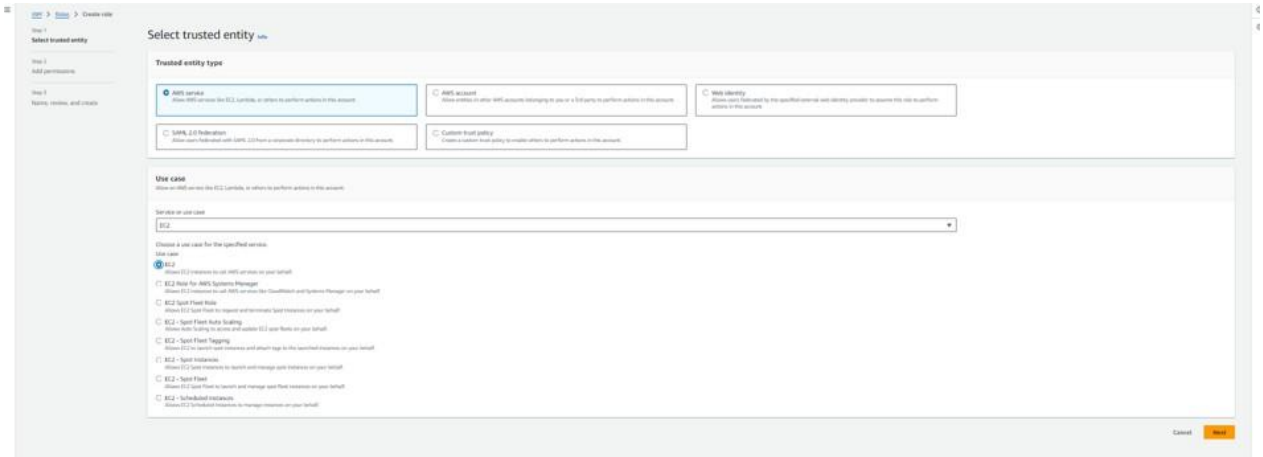
Description: Start the Flask application on all network interfaces at port 80 with debug mode enabled for easier development and testing.

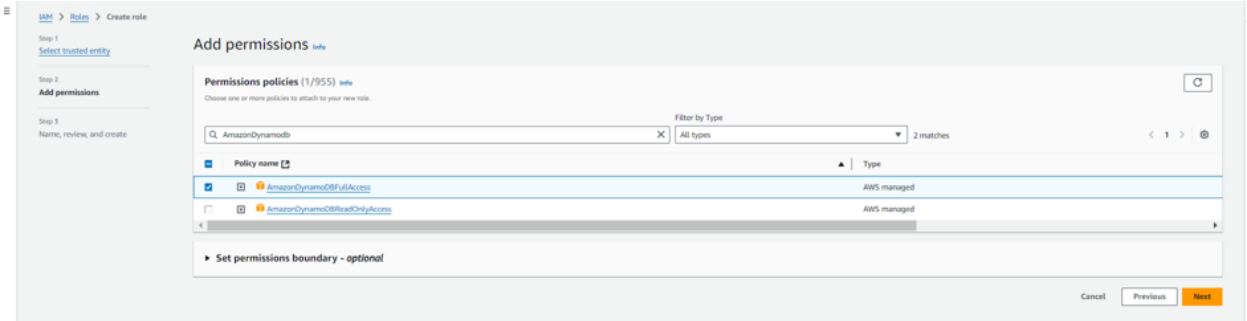
Milestone 5: IAM Role Setup

- **Activity 5.1:Create IAM Role.**
    - In the AWS Console, go to IAM and create a new IAM Role for EC2 to interact with DynamoDB and SNS.





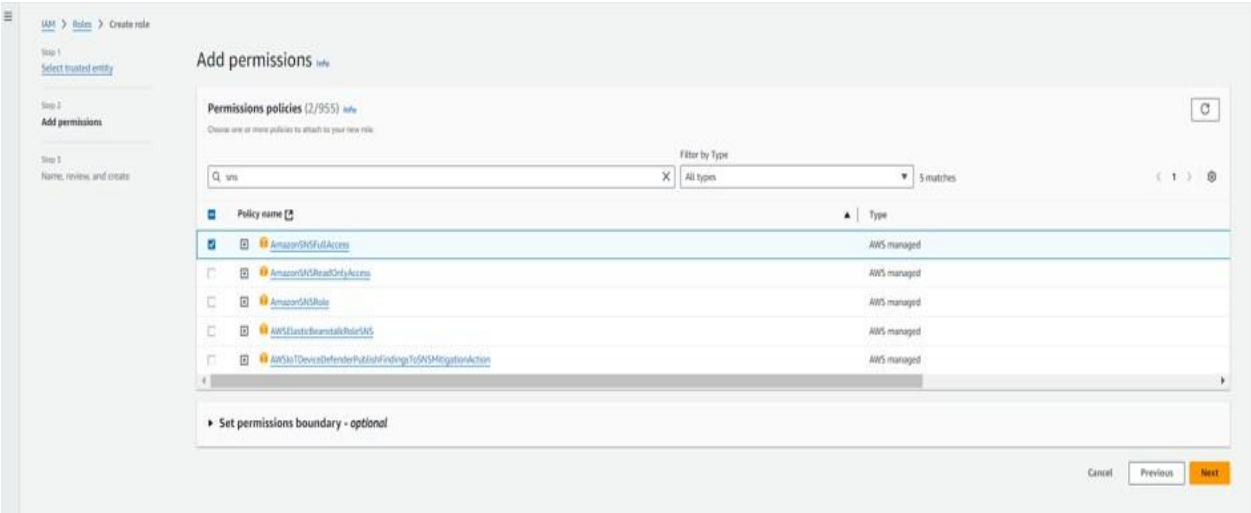


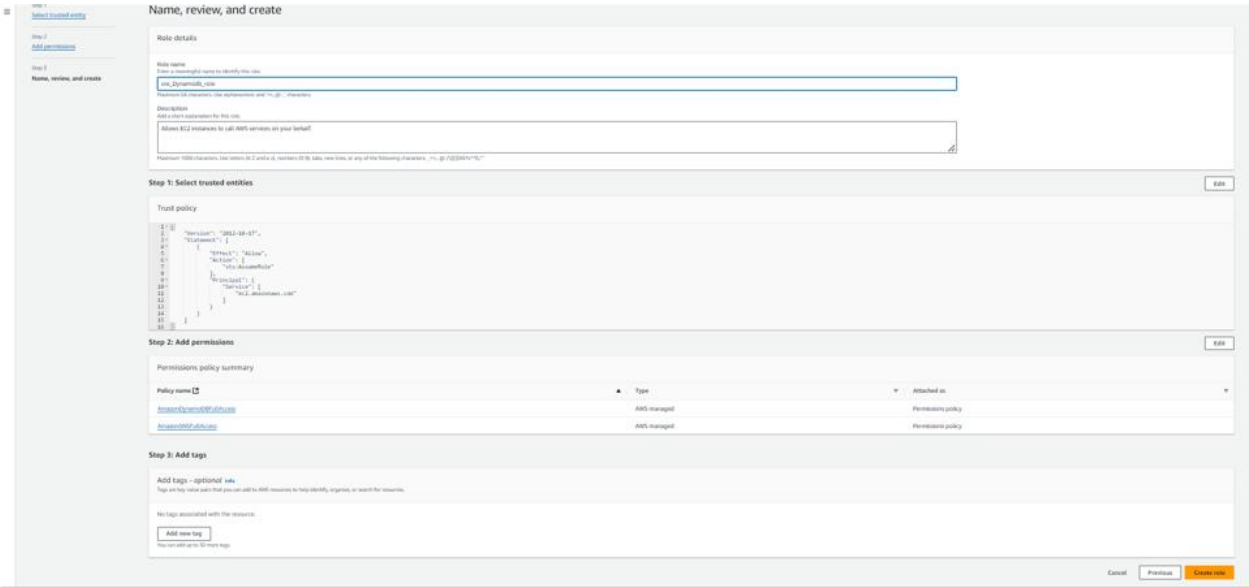


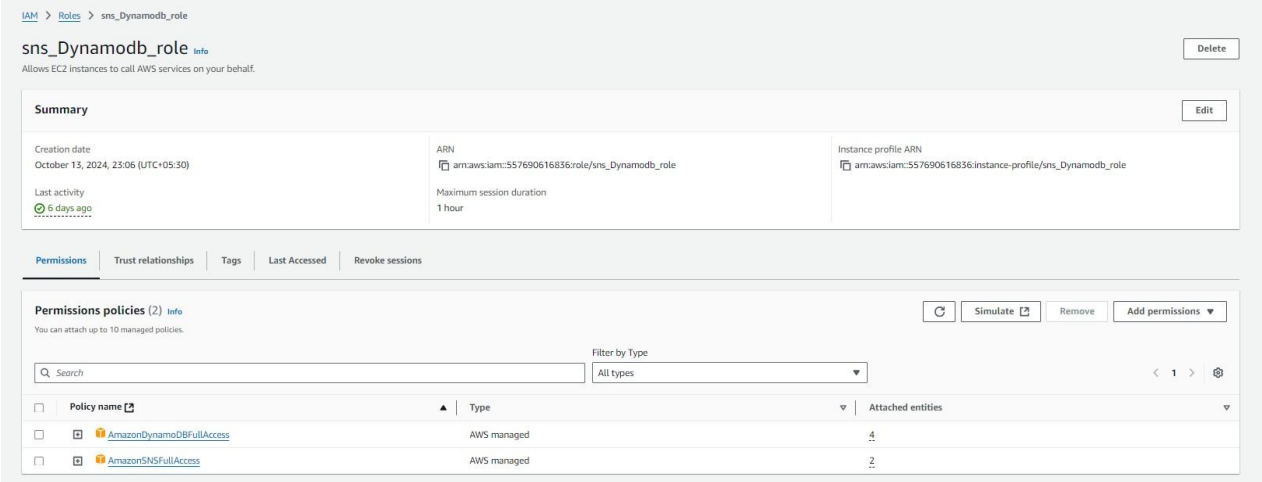
- **Activity 5.2: Attach Policies.**

  Attach the following policies to the role:

  - AmazonDynamoDBFullAccess: Allows EC2 to perform read/write operations on DynamoDB.
  - AmazonSNSFullAccess: Grants EC2 the ability to send notifications via SNS.

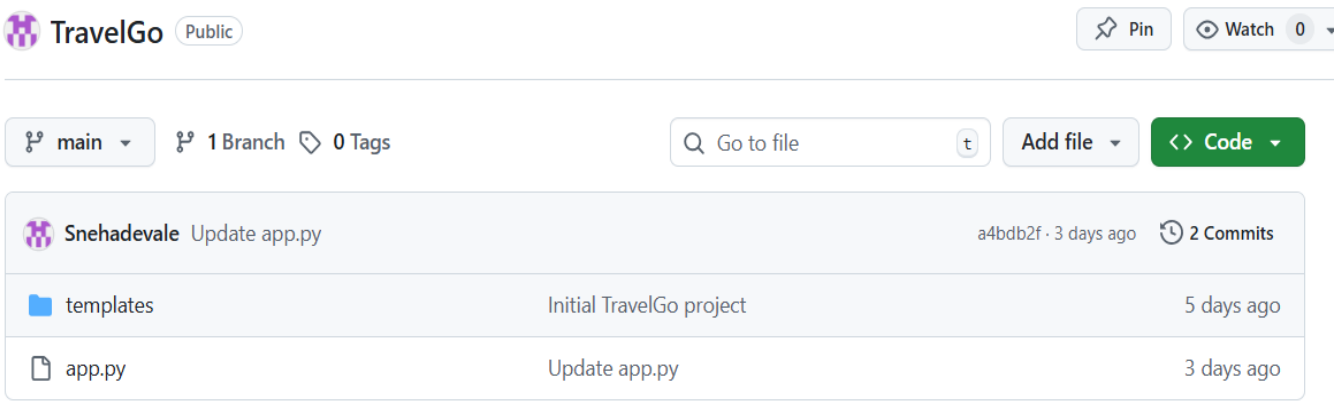


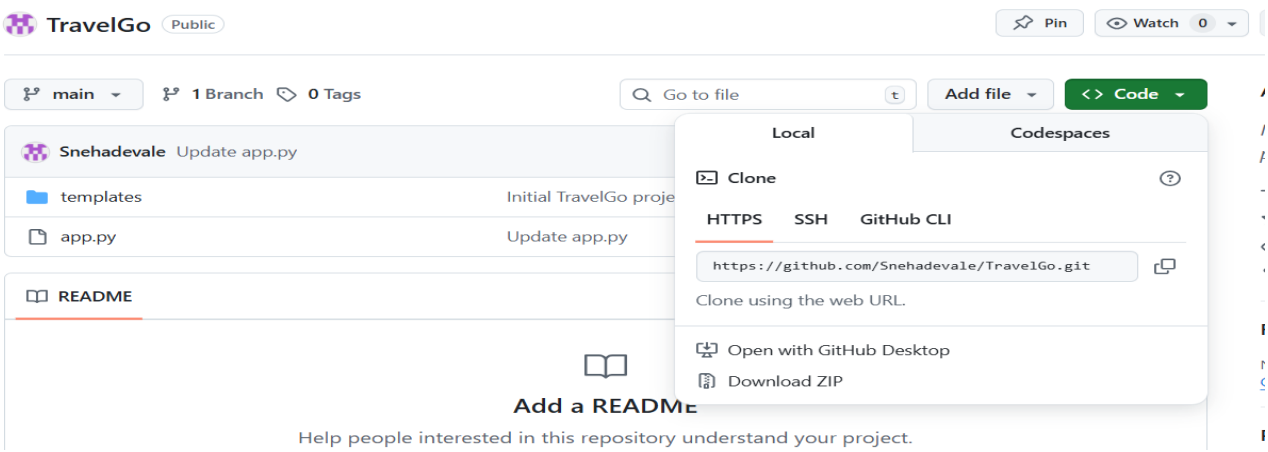




Milestone 6: EC2 Instance Setup

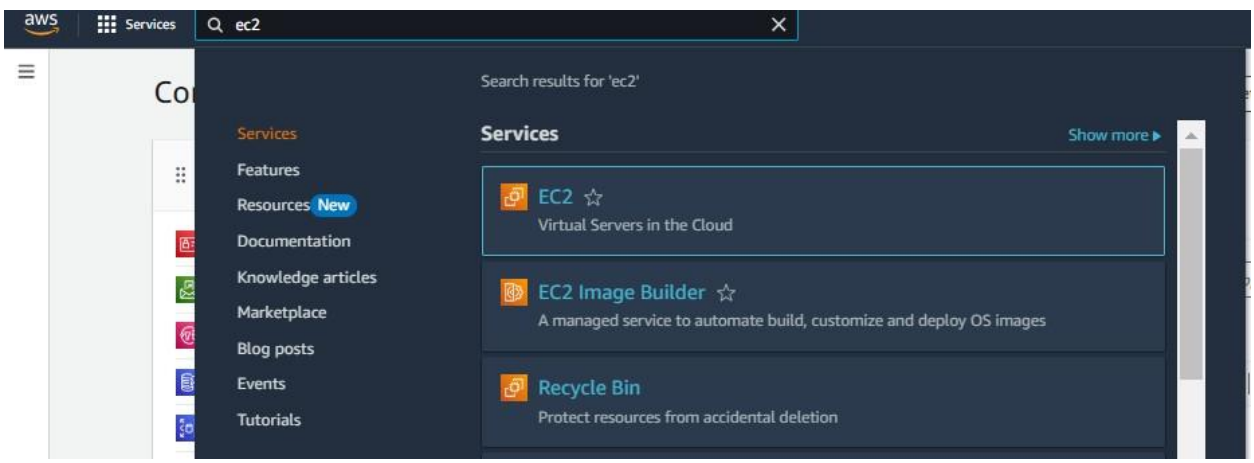
- **Note: Load your Flask app and Html files into GitHub repository.**



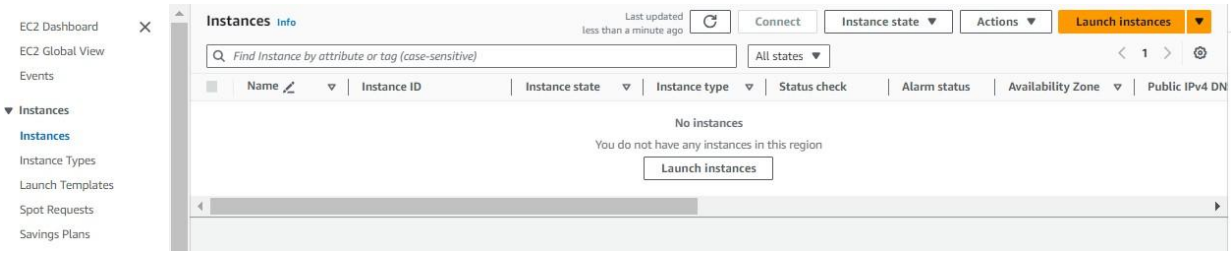


- **Activity 6.1: Launch an EC2 instance to host the Flask application.**

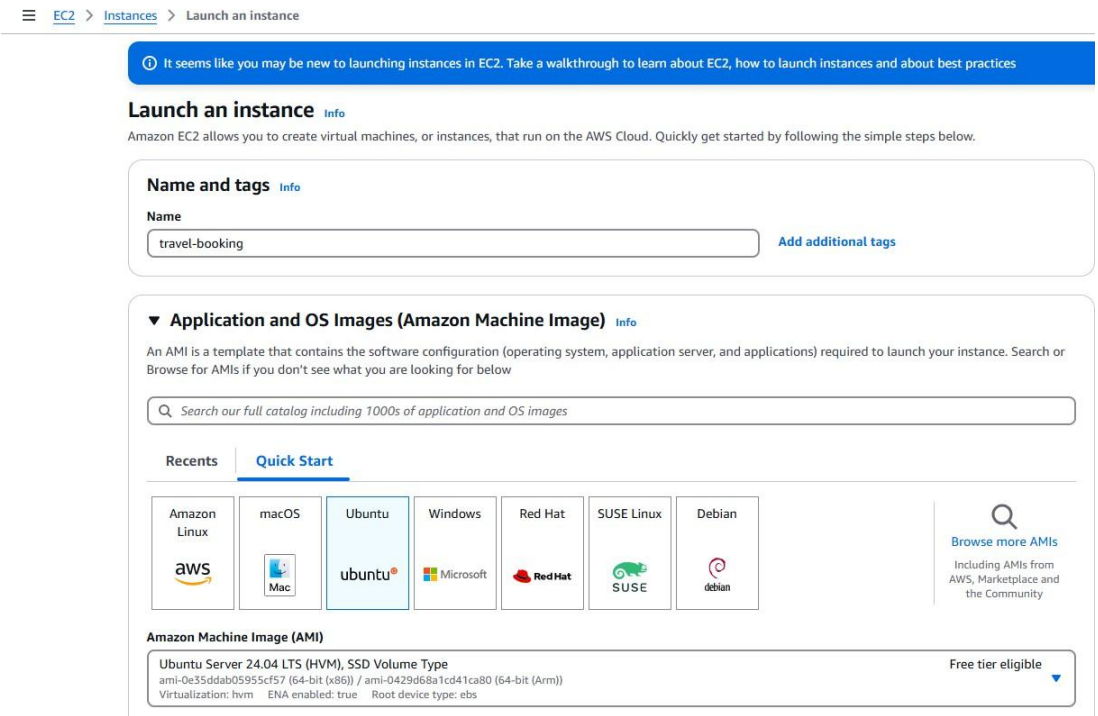
  - **Launch EC2 Instance**
    - In the AWS Console, navigate to EC2 and launch a new instance.



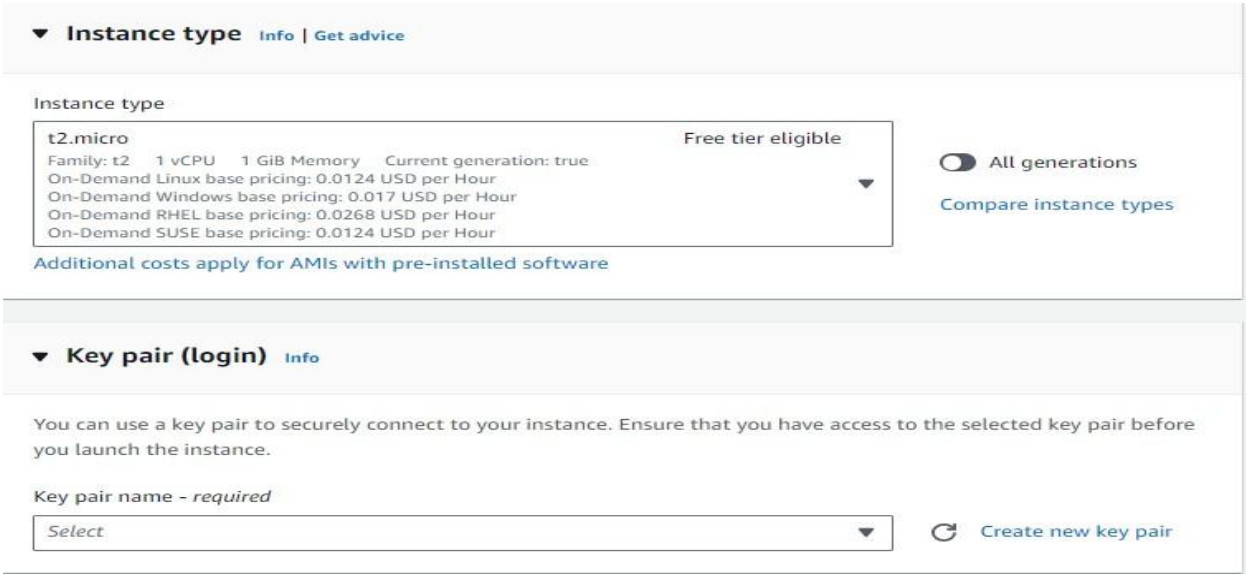
- Click on Launch instance to launch EC2 instance

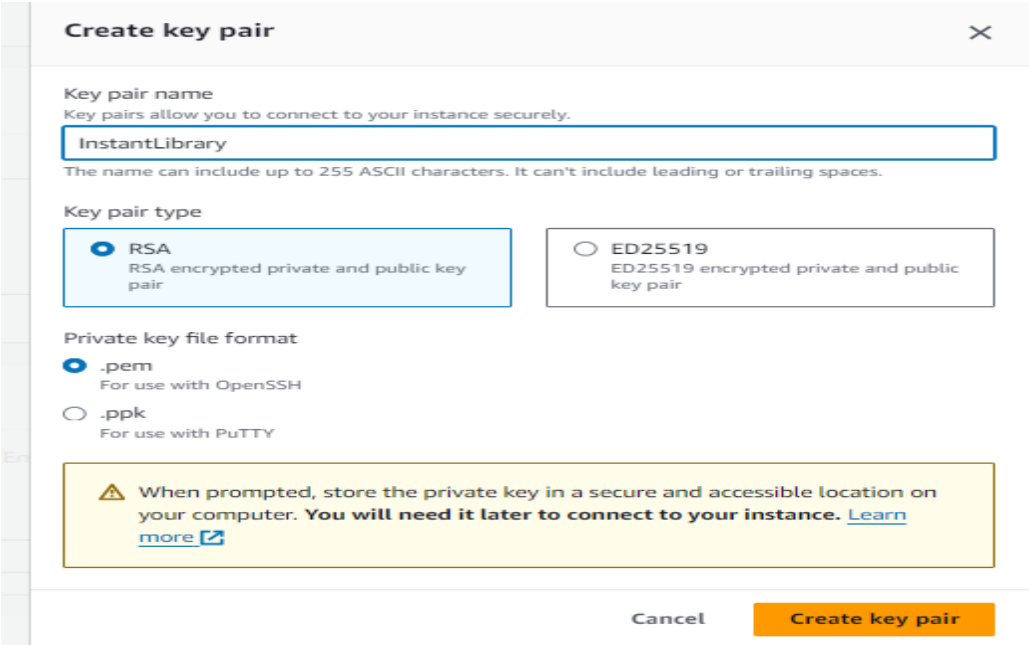


- Choose Amazon Linux 2 or Ubuntu as the AMI and t2.micro as the instance type (free-tier eligible).

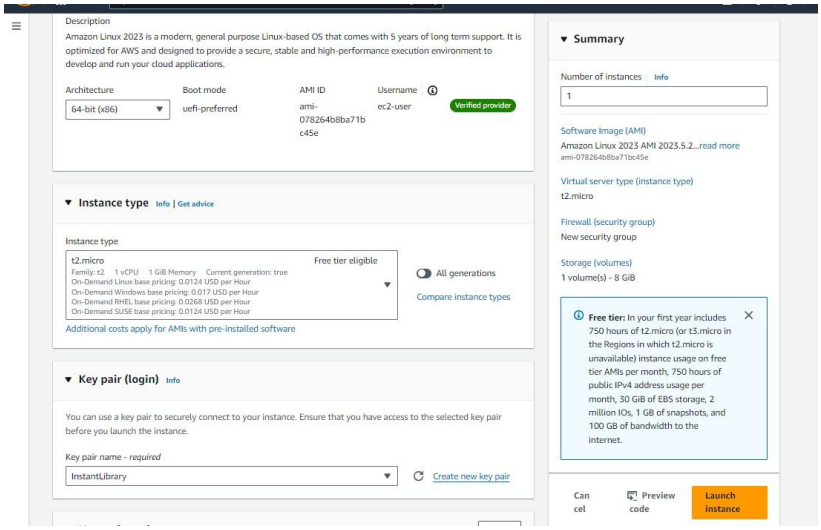


- Create and download the key pair for Server access.

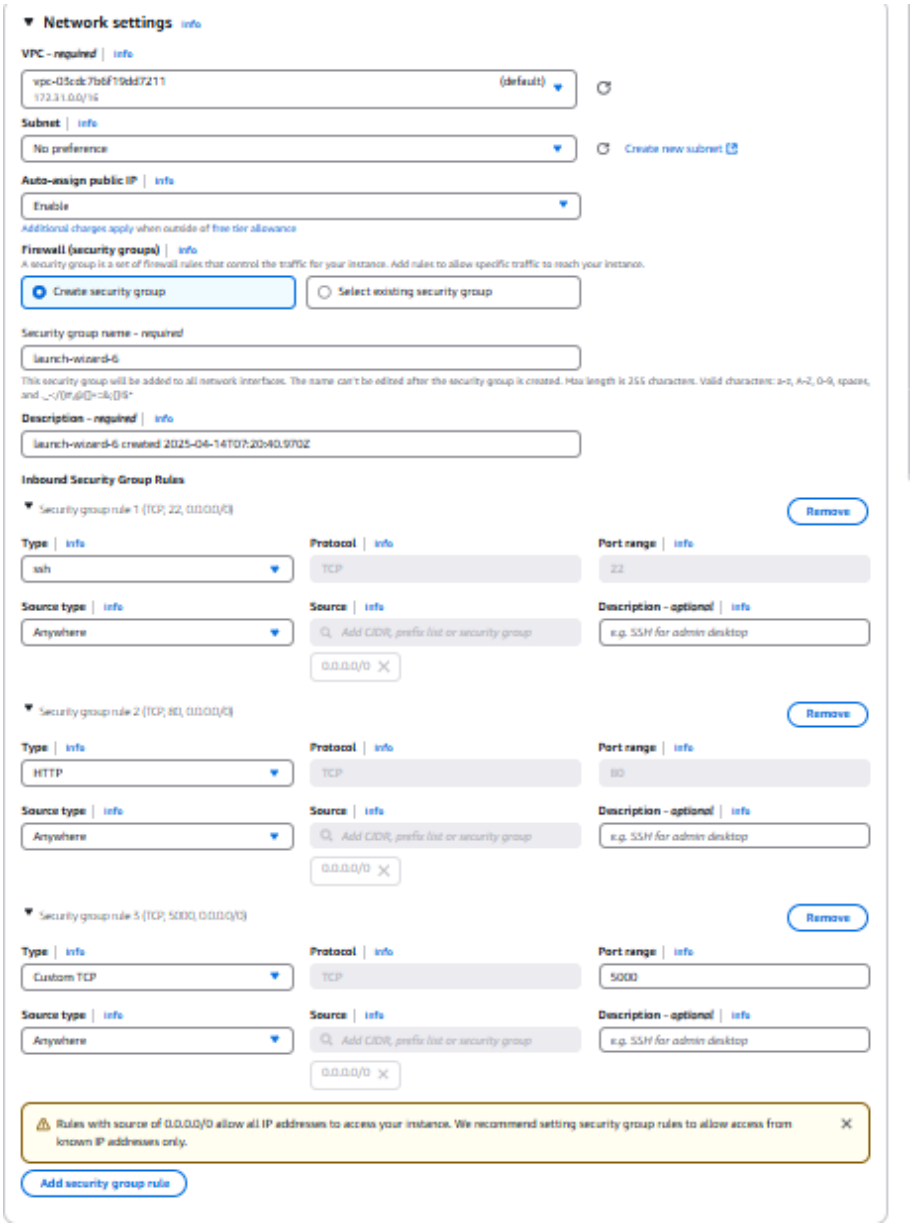


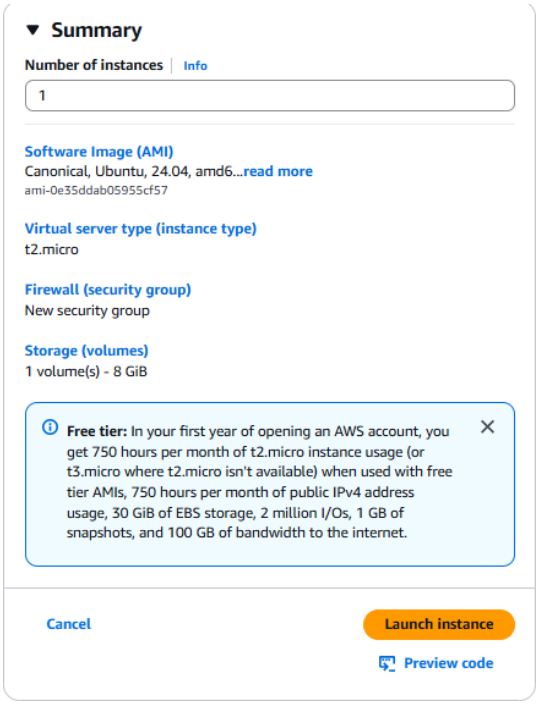

Create key pair
Key pair name
Key pairs allow you to connect to your instance securely.
InstantLibrary
The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.
Key pair type
RSA
RSA encrypted private and public key pair
ED25519
ED25519 encrypted private and public key pair
Private key file format
.pem
For use with OpenSSH
.ppk
For use with PuTTY
When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. Learn more
Cancel
Create key pair

InstantLibrary.pem

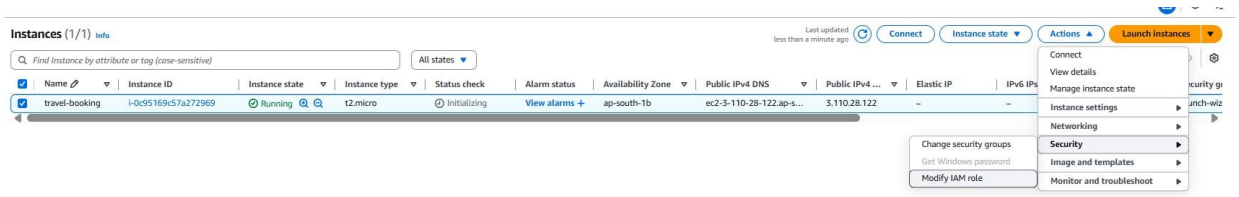

Description
Amazon Linux 2023 is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.
Architecture
64-bit (x86)
Boot mode
uefi-preferred
AMI ID
ami-078264b8ba71bc45e
Username
ec2-user
Verified provider
Instance type
Info | Get advice
Instance type
t2.micro
Free tier eligible
Family: t2 1 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0124 USD per Hour
On-Demand Windows base pricing: 0.017 USD per Hour
On-Demand RHEL base pricing: 0.0268 USD per Hour
On-Demand SUSE base pricing: 0.0124 USD per Hour
All generations
Compare instance types
Additional costs apply for AMIs with pre-installed software
Key pair (login)
Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
InstantLibrary
Create new key pair
Summary
Number of instances
Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.5.2...read more
ami-078264b8ba71bc45e
Virtual server type (instance type)
t2.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 750 hours of public IPv4 address usage per month, 30 GiB of EBS storage, 2 million IOs, 1 GB of snapshots, and 100 GB of bandwidth to the internet.
Cancel
Preview code
Launch instance

- **Activity 6.2: Configure security groups for HTTP, and SSH access.**

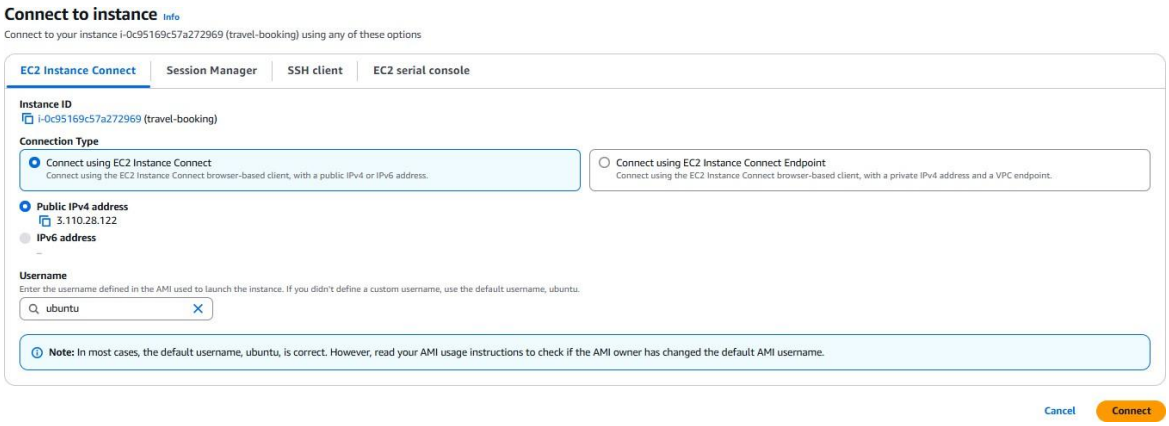


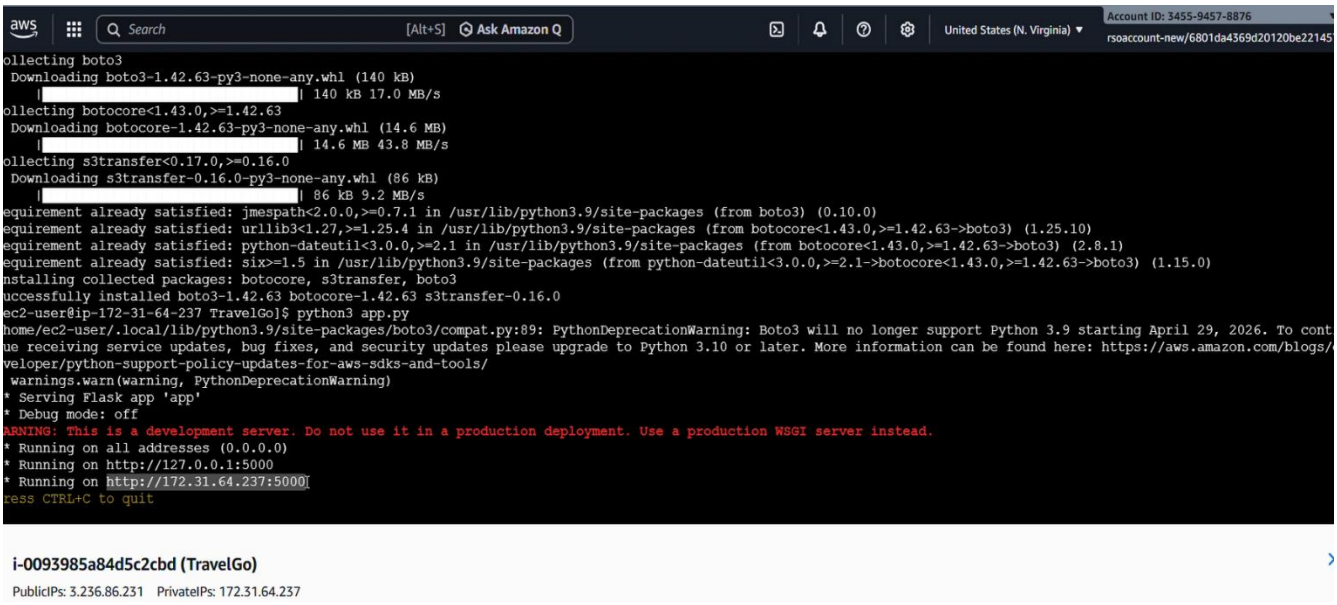


- To connect to EC2 using **EC2 Instance Connect**, start by ensuring that an **IAM role** is attached to your EC2 instance. You can do this by selecting your instance, clicking on **Actions**, then navigating to **Security** and selecting **Modify IAM Role** to attach the appropriate role. After the IAM role is connected, navigate to the **EC2** section in the **AWS Management Console**. Select the **EC2 instance** you wish to connect to. At the top of the **EC2 Dashboard**, click the **Connect** button. From the connection methods presented, choose **EC2 Instance Connect**. Finally, click **Connect** again, and a new browser-based terminal will open, allowing you to access your EC2 instance directly from your browser.



- Now connect the EC2 with the files





Milestone 7: Deployment on EC2

Activity 7.1: Install Software on the EC2 Instance

 # Update and install Python + pip

sudo apt update

sudo apt install python3-pip -y

sudo apt install git -y

sudo apt install python3-venv -y

python3 -m venv venv

source venv/bin/activate

Activity 7.2:Clone Your Flask Project from GitHub

Clone your project repository from GitHub into the EC2 instance using Git.

Run: 'git clone https://github.com/your-github-username/your-repository-name.git'

Note: change your-github-username and your-repository-name with your credentials

here: 'https://github.com/Snehadevale/TravelGo.git'
This will download your project to the EC2 instance.

To navigate to the project directory, run the following command:

cd TravelGo

Once inside the project directory, configure and run the Flask application by executing the following command with elevated privileges:

Run the Flask Application

export SNS_TOPIC_ARN=arn:aws:sns:ap-south-1:557690616836:BookingConfirmation

git pull origin main (Only if you made changes in git and want to reflect them in the EC2 terminal.)

 # Install requirements

pip install flask boto3

#Run Flask app

Python3 app.py

Launch the Flask app:

sudo -E venv/bin/python3 app.py

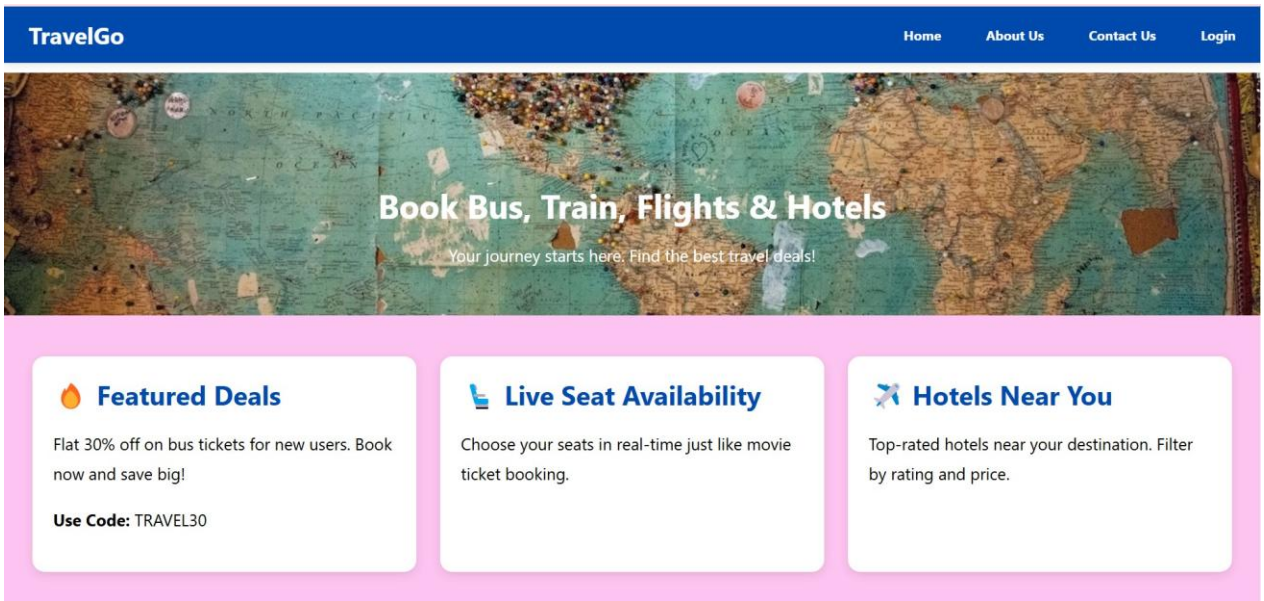
Access the website through:

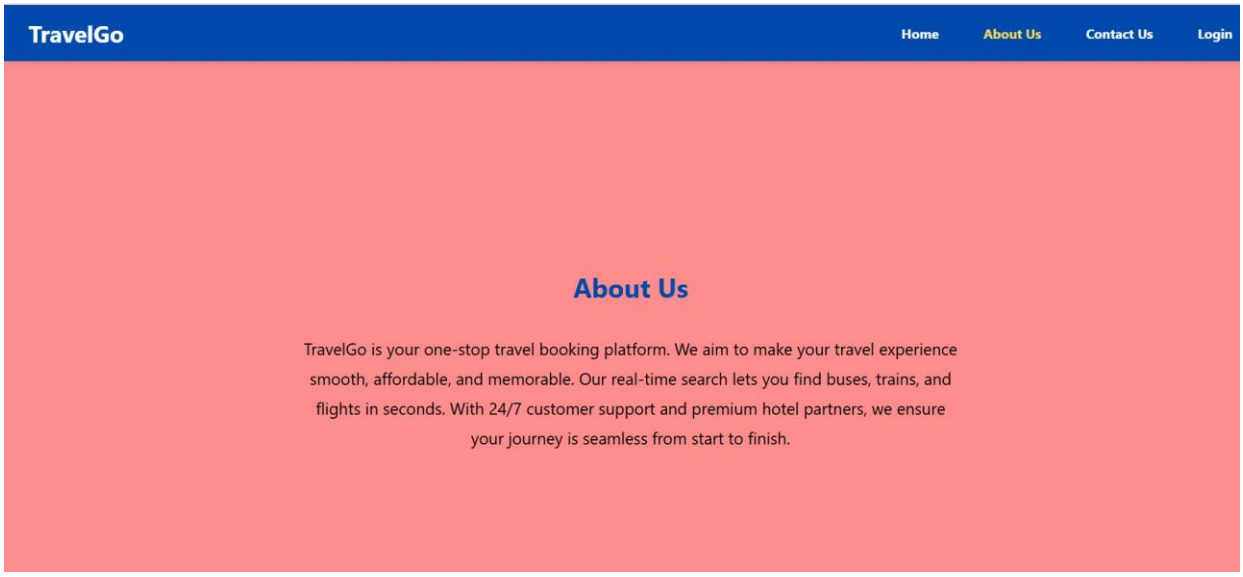
PublicIPs: http://3.236.86.231/

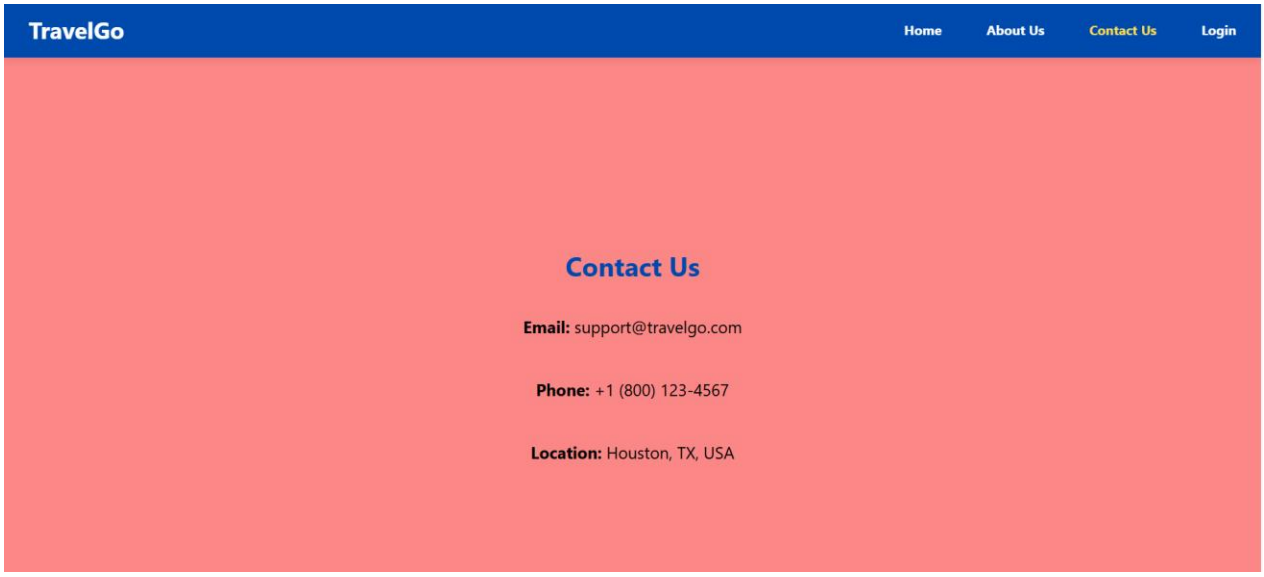
Milestone 8: Testing and Deployment

- **Activity 8.1: Conduct functional testing to verify user registration, login, book requests, and notifications.**

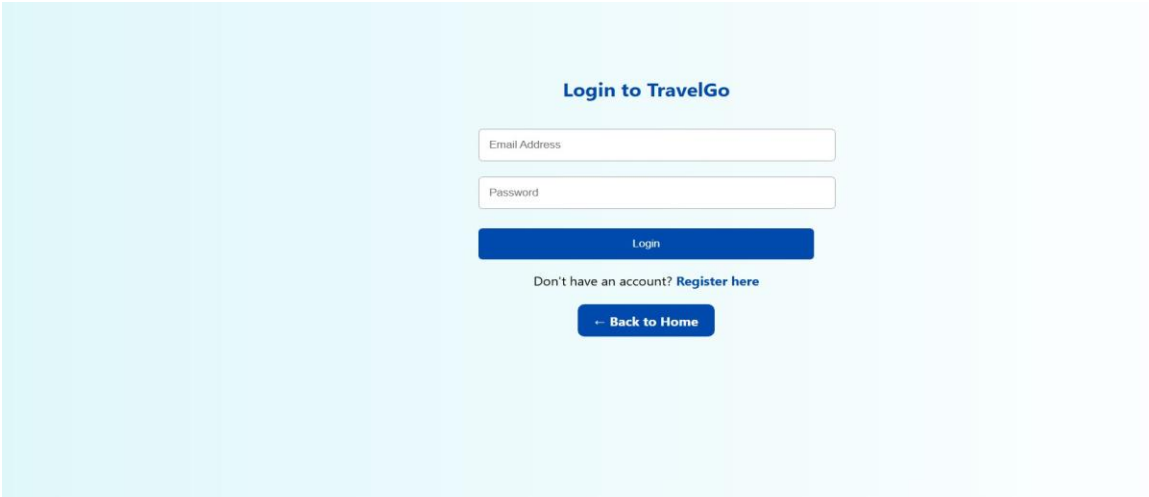
Home Page:



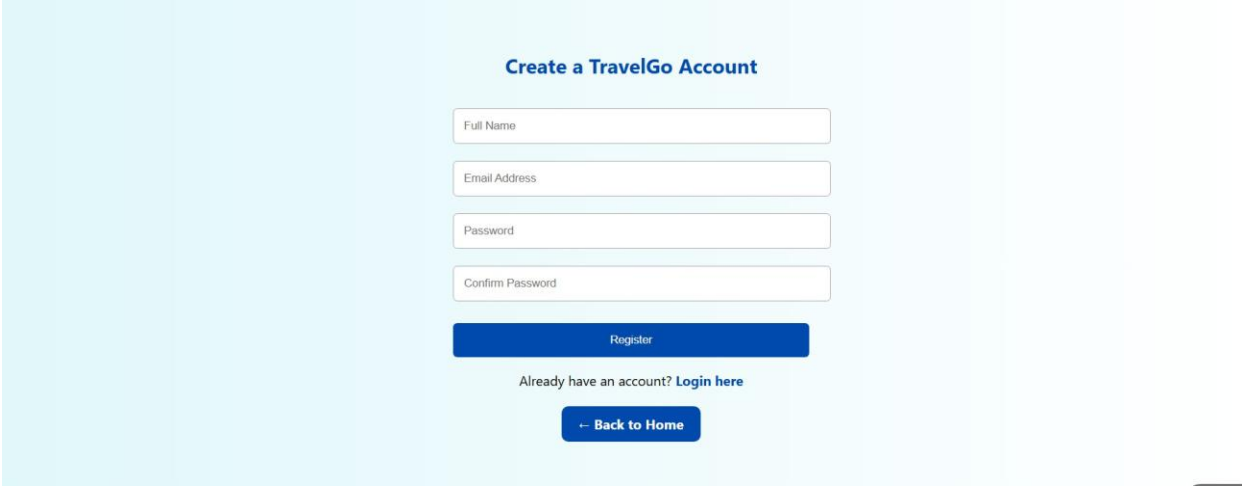



TravelGo
Home
About Us
Contact Us
Login
Contact Us
Email: support@travelgo.com
Phone: +1 (800) 123-4567
Location: Houston, TX, USA

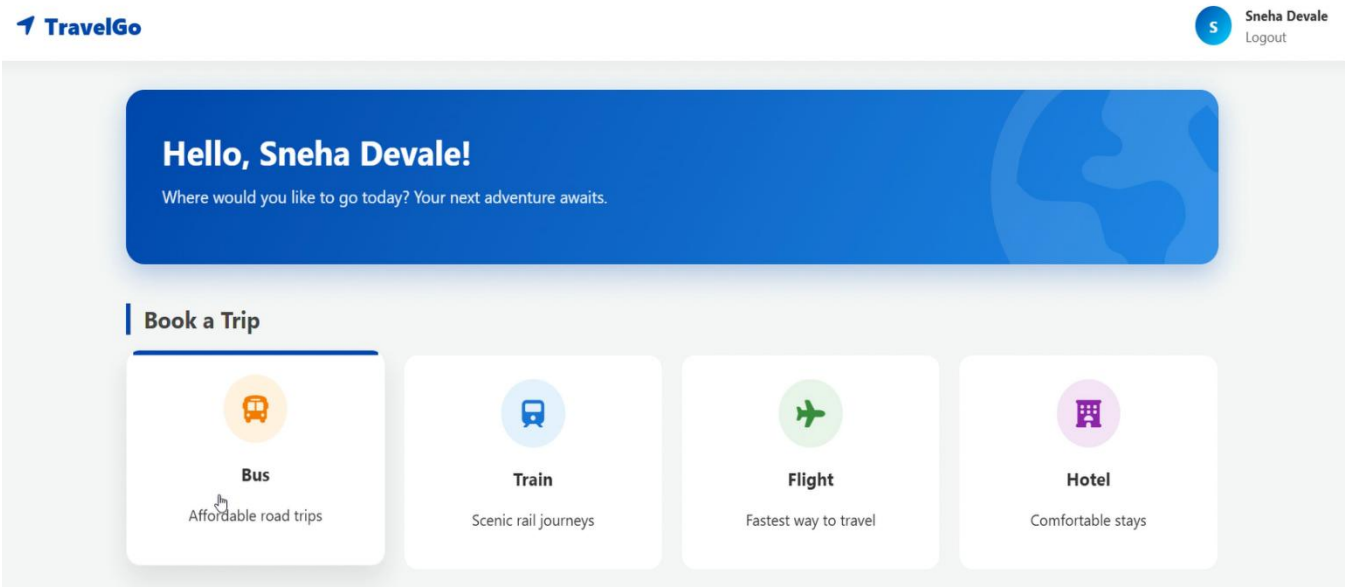
Login Page:

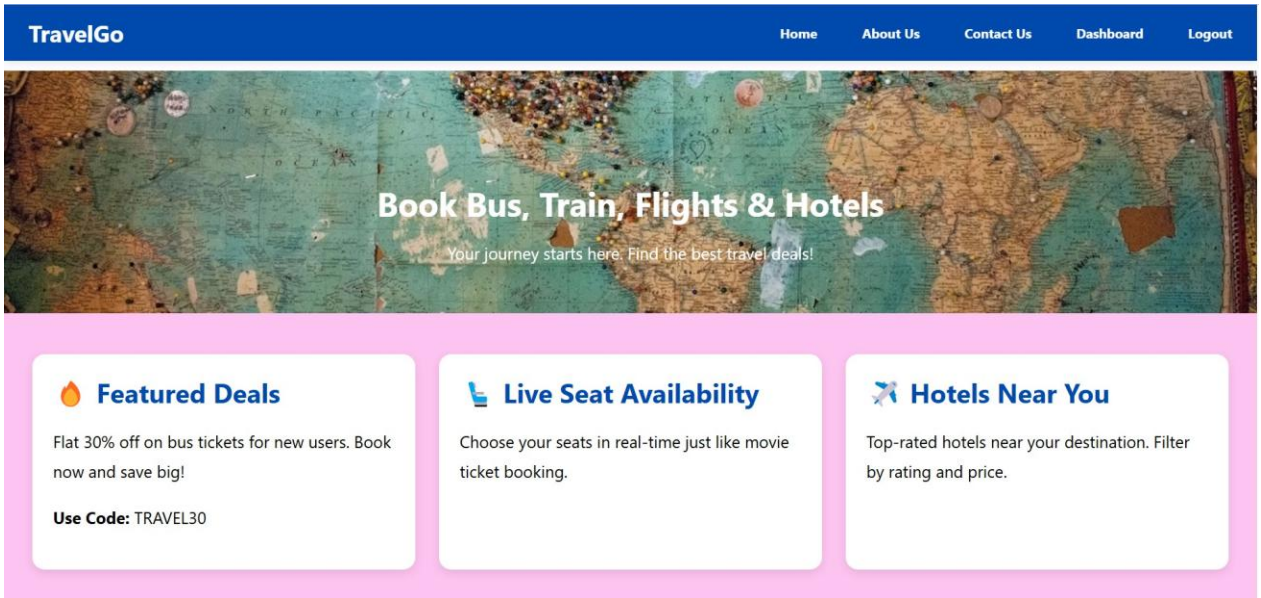

Login to TravelGo
Email Address
Password
Login
Don't have an account? Register here
← Back to Home

Register Page:

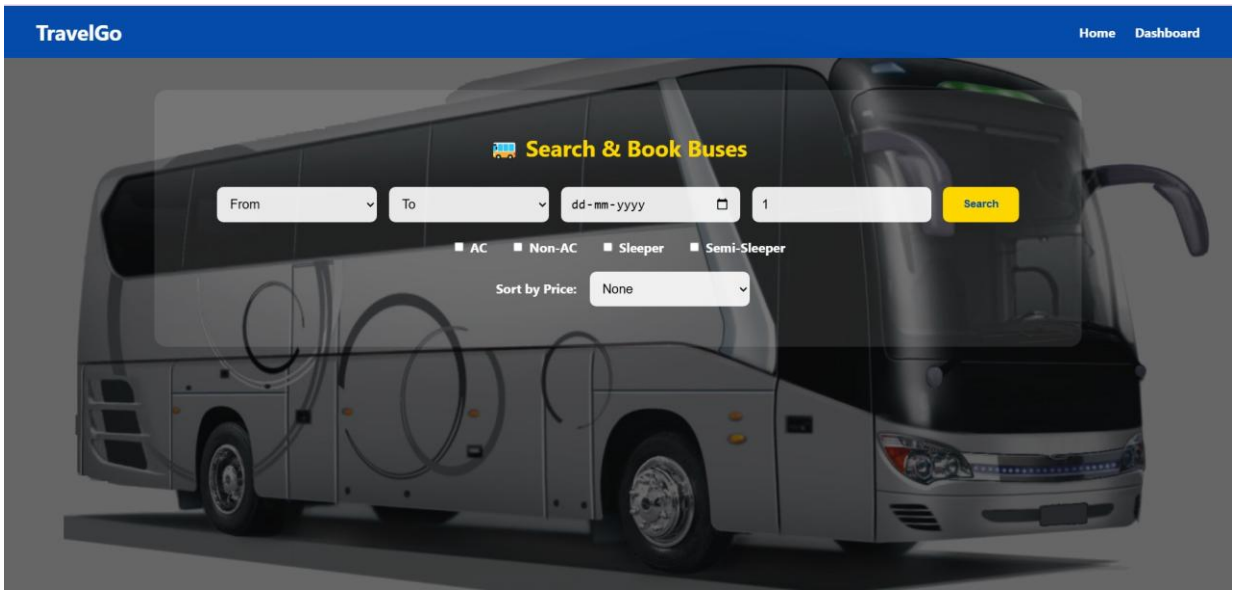


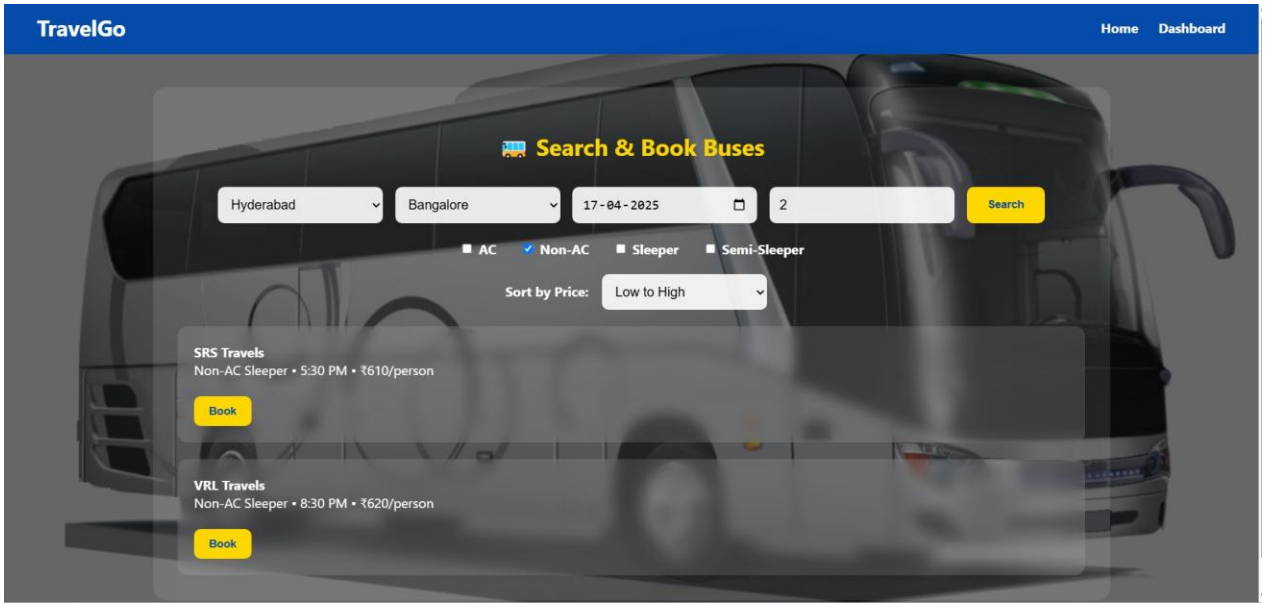
Dashboard page:

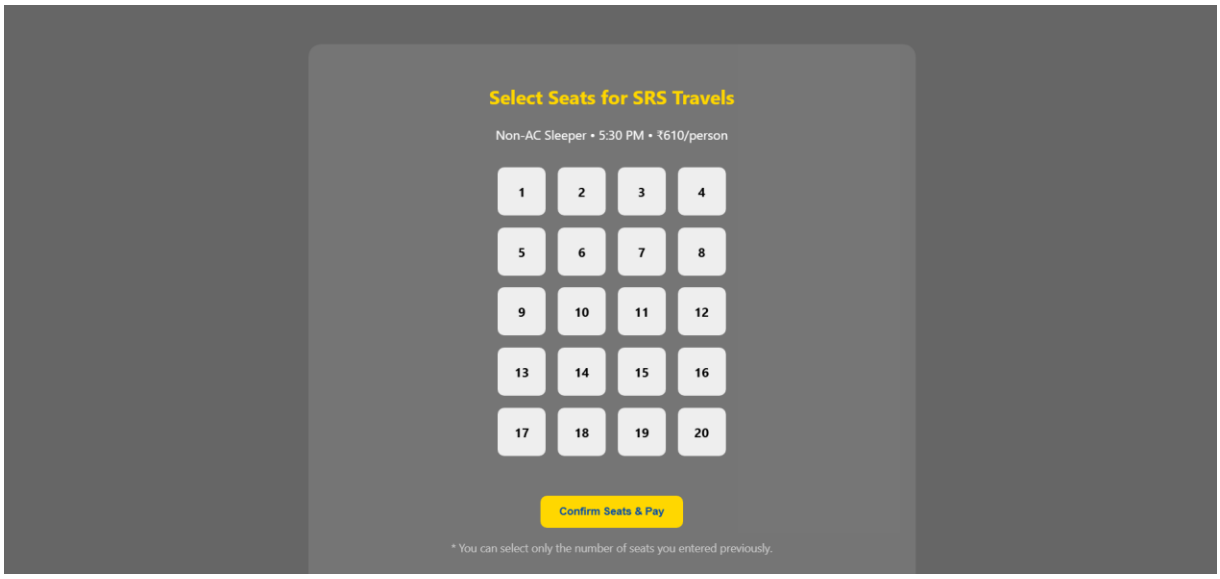


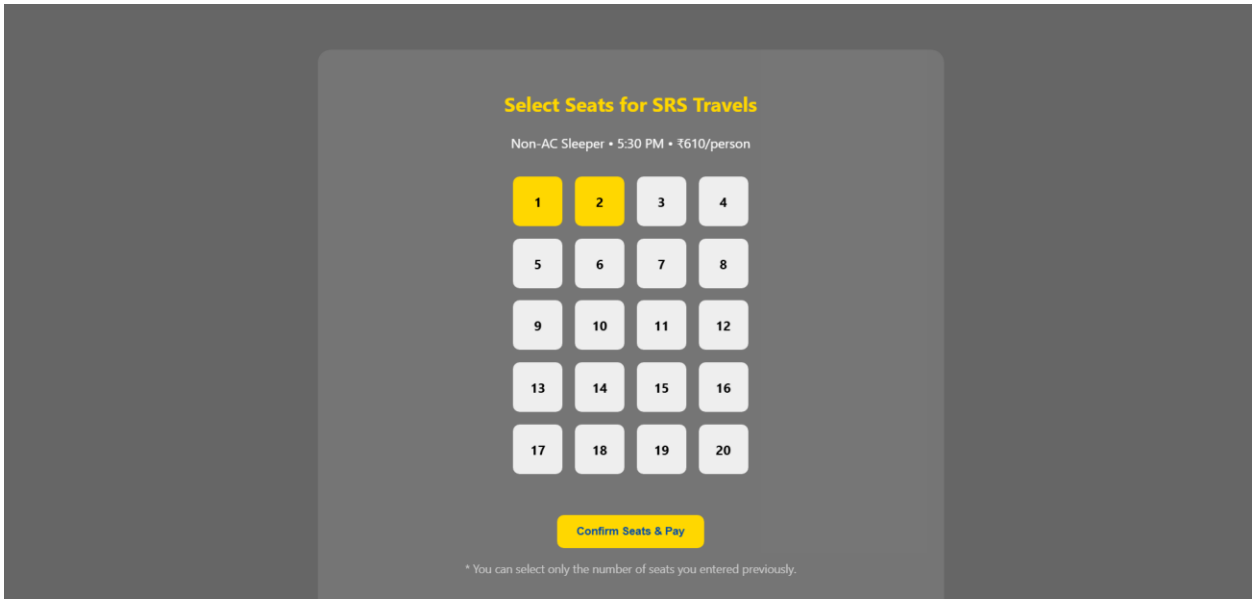

TravelGo
Home
About Us
Contact Us
Dashboard
Logout
Book Bus, Train, Flights & Hotels
Your journey starts here. Find the best travel deals!
Featured Deals
Flat 30% off on bus tickets for new users. Book now and save big!
Use Code: TRAVEL30
Live Seat Availability
Choose your seats in real-time just like movie ticket booking.
Hotels Near You
Top-rated hotels near your destination. Filter by rating and price.

Buses page:

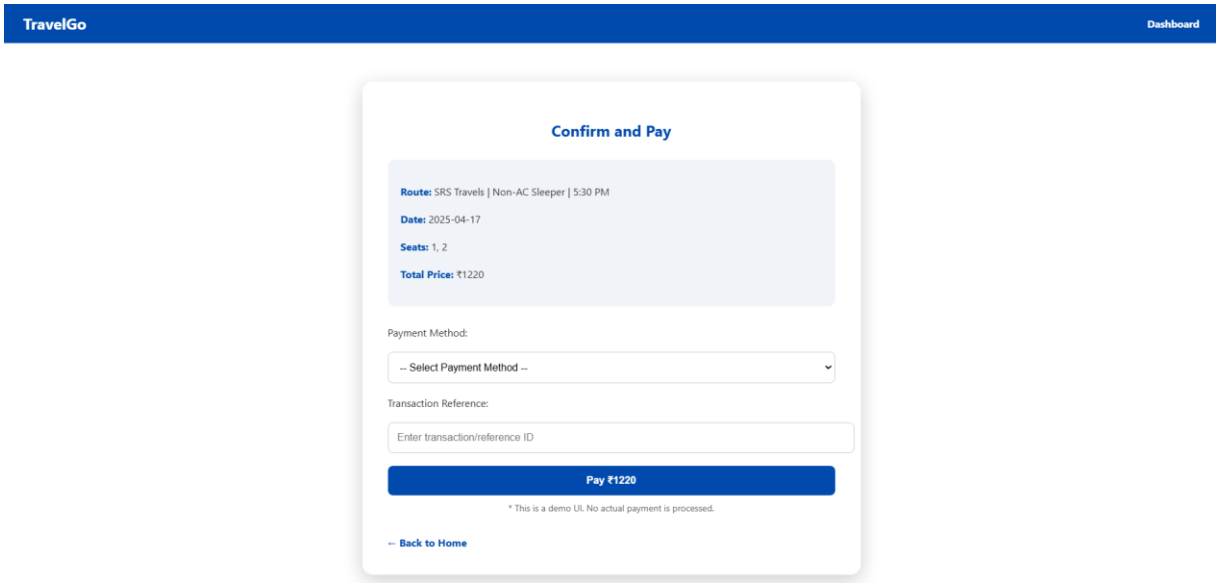

TravelGo
Home
Dashboard
Search & Book Buses
From
To
dd-mm-yyyy
1
Search
AC
Non-AC
Sleeper
Semi-Sleeper
Sort by Price:
None


TravelGo
Home
Dashboard
Search & Book Buses
Hyderabad
Bangalore
17-04-2025
2
Search
AC
Non-AC
Sleeper
Semi-Sleeper
Sort by Price:
Low to High
SRS Travels
Non-AC Sleeper • 5:30 PM • ₹610/person
Book
VRL Travels
Non-AC Sleeper • 8:30 PM • ₹620/person
Book

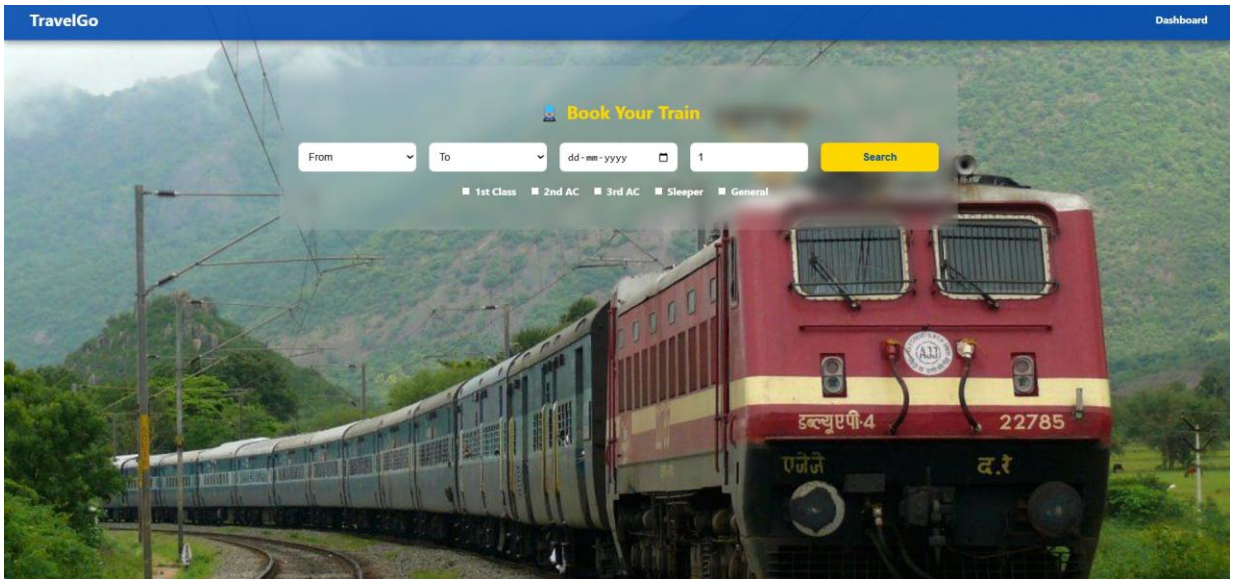

Select Seats for SRS Travels
Non-AC Sleeper • 5:30 PM • ₹610/person
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
Confirm Seats & Pay
* You can select only the number of seats you entered previously.

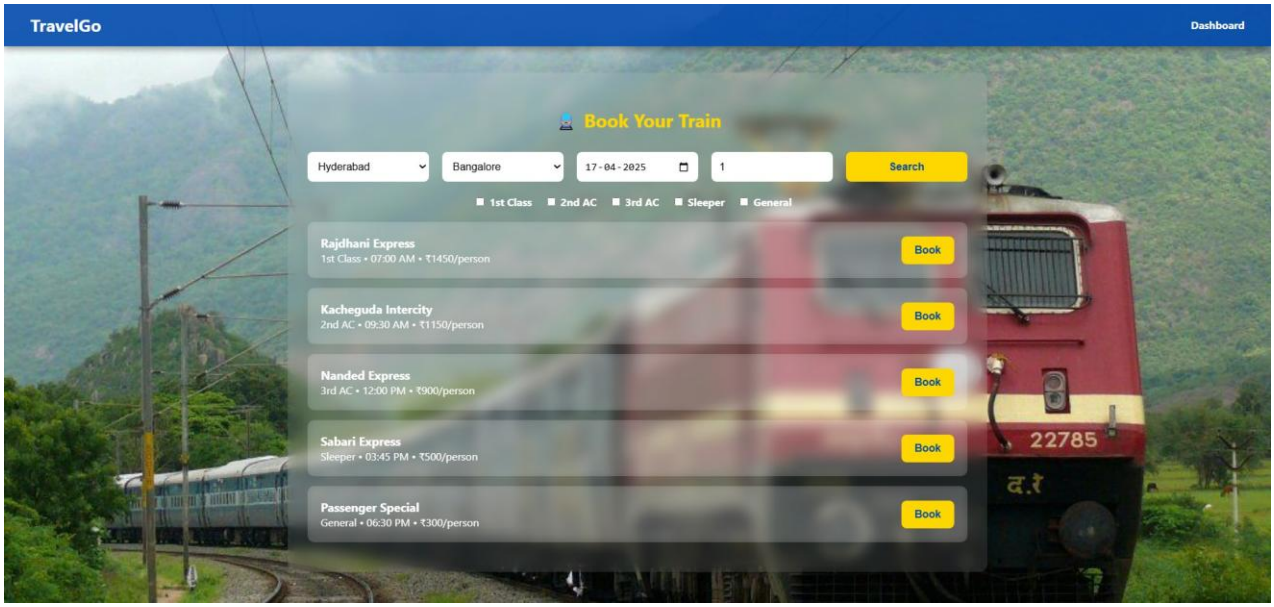


Payment Page:

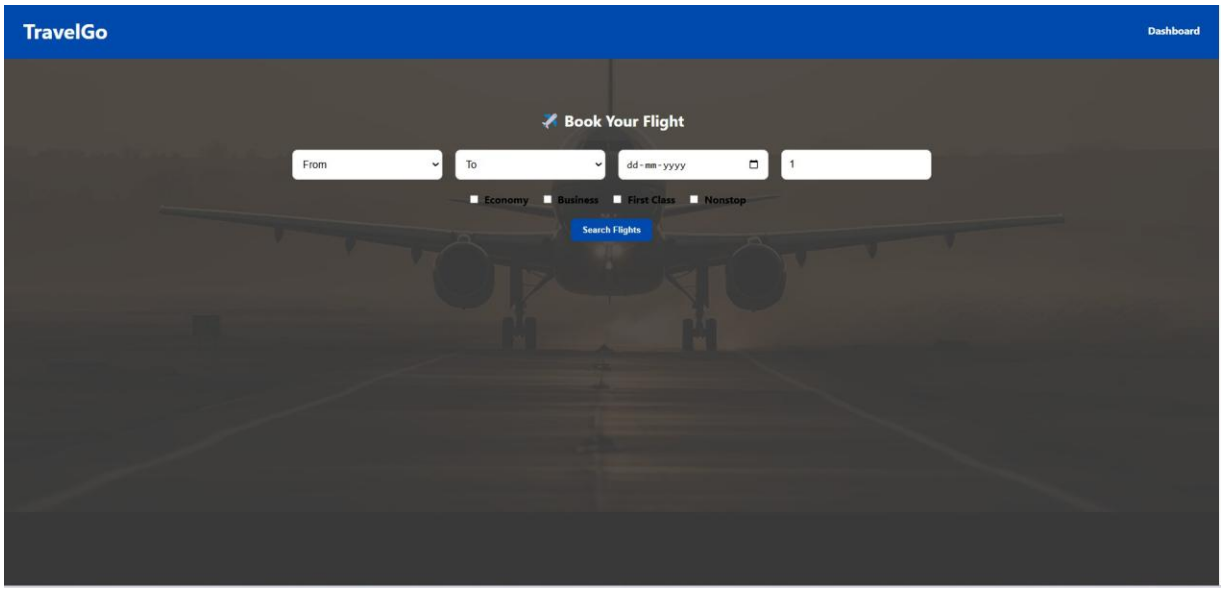


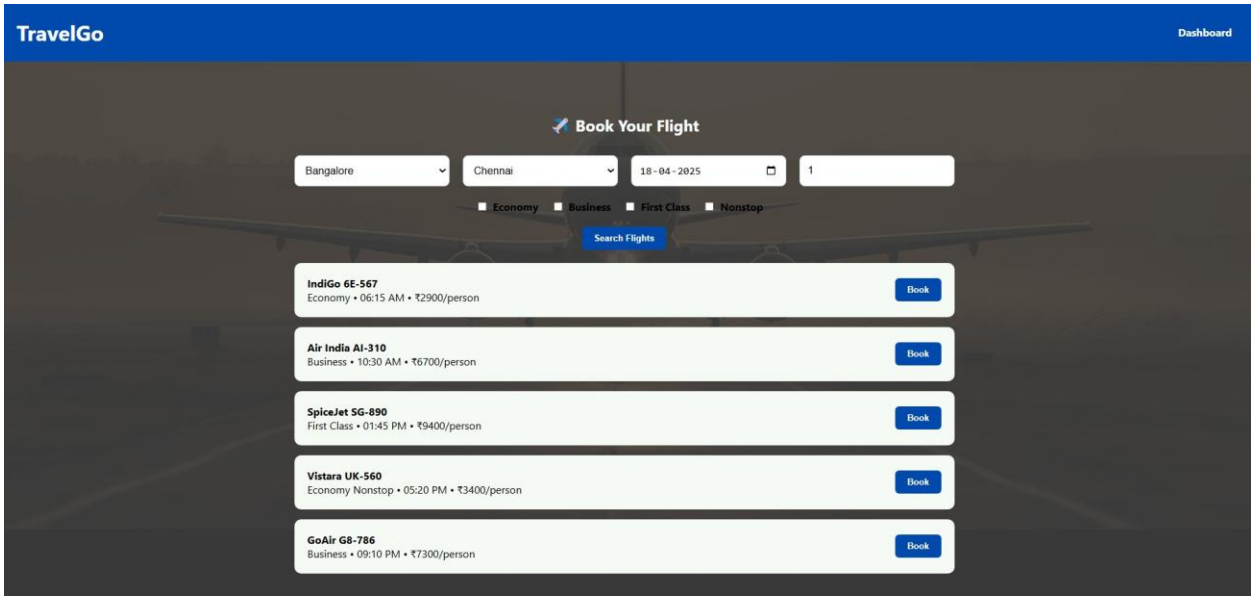
Trains page:


TravelGo
Dashboard
Book Your Train
From
To
dd-mm-yyyy
1
Search
1st Class
2nd AC
3rd AC
Sleeper
General
डब्ल्यूएपी-4
22785

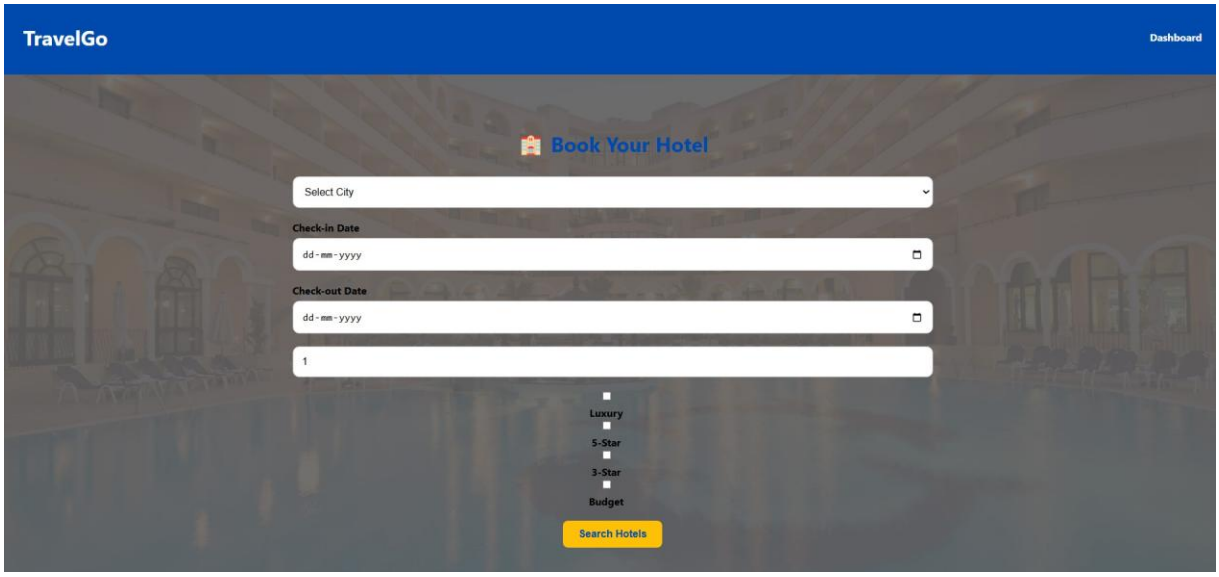


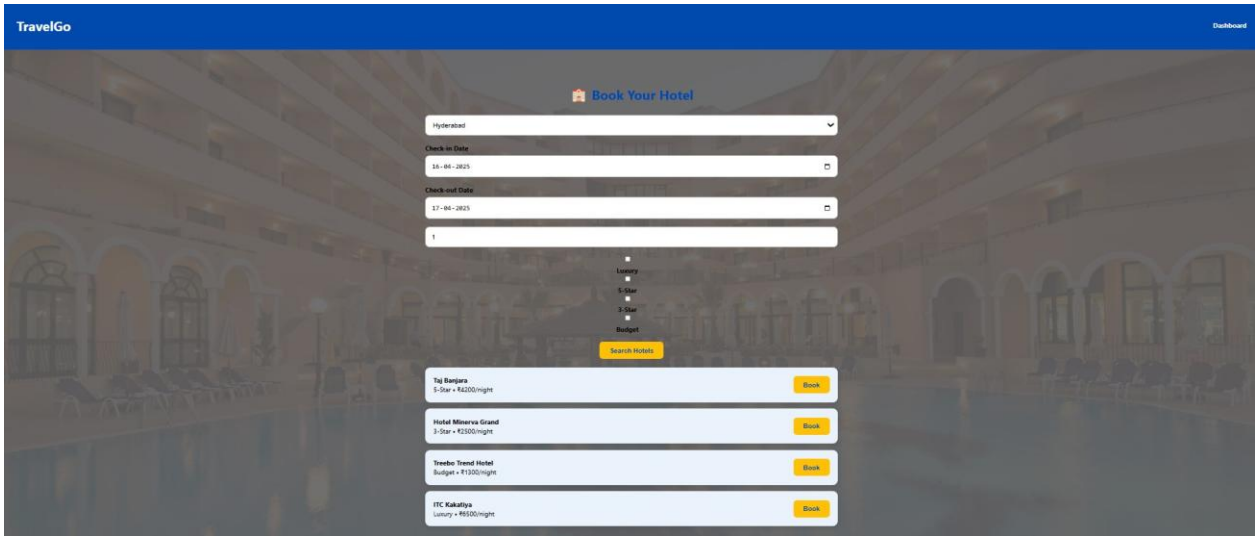
Flights Page:




TravelGo
Dashboard
Book Your Flight
Bangalore
Chennai
18-04-2025
1
Economy
Business
First Class
Nonstop
Search Flights
IndiGo 6E-567
Economy • 06:15 AM • ₹2900/person
Book
Air India AI-310
Business • 10:30 AM • ₹6700/person
Book
SpiceJet SG-890
First Class • 01:45 PM • ₹9400/person
Book
Vistara UK-560
Economy Nonstop • 05:20 PM • ₹3400/person
Book
GoAir G8-786
Business • 09:10 PM • ₹7300/person
Book

Hotels Page:


TravelGo
Dashboard
Book Your Hotel
Select City
Check-in Date
dd-mm-yyyy
Check-out Date
dd-mm-yyyy
1
Luxury
5-Star
3-Star
Budget
Search Hotels


TravelGo
Dashboard
Book Your Hotel
Hyderabad
Check-in Date
16-04-2025
Check-out Date
17-04-2025
1
Luxury
5-Star
3-Star
Budget
Search Hotels
Taj Banjara
5-Star • ₹4200/night
Book
Hotel Minerva Grand
3-Star • ₹2500/night
Book
Treebo Trend Hotel
Budget • ₹1300/night
Book
ITC Kakatiya
Luxury • ₹6500/night
Book

Dynamodb Database updations :

1. **travel-users table :**

Table: travel-Users - Items returned (3)

Scan started on April 18, 2025, 11:21:41

Actions ▼ Create item

1

email *(String)*	logins	name	password
kilarukusuma12@gm...	1	KILARU KU...	123
sunder@thesmartbrid...	1	Sunder	12345
alekhya080228@gma...	3	P_Alekhya	1234

Description: Store and manage user credentials, login counts, and names in the `travel-Users` table on DynamoDB for authentication and tracking purposes.

2. **bookings table :**

Table: Bookings - Items returned (5)

Scan started on April 18, 2025, 11:22:33

Actions ▼ Create item

1

email *(String)*	booking_id *(String)*	date	destination	details	payment_met...	payment_reference	p
kilarukusuma12@gm...	7bc700fe	2025-04-26	Chennai	Air India AI-...	Net Banking	123456	4
alekhya080228@gma...	5af93005	2025-04-1...	Hotel Minerva...	Hotel Miner...	Wallet	1234	2
alekhya080228@gma...	6026a7a6	2025-04-17	Bangalore	SRS Travels ...	UPI	12345	1
alekhya08022...	965e4155	2025-04-17	Bangalore	Rajdhani Ex...	UPI	1234	1
alekhya080228@gma...	b0a35b1c	2025-04-18	Chennai	IndiGo 6E-5...	Net Banking	123456	2

Description: Use the Bookings table in DynamoDB to store user travel bookings including date, destination, transport details, and payment information for each transaction.

Conclusion:

The TravelGo platform has been successfully designed and implemented as a cloud-based travel booking system that simplifies the process of reserving transportation and accommodation services. By integrating modern web development technologies with cloud infrastructure, the project demonstrates how travel services such as bus, train, flight, and hotel bookings can be managed efficiently through a single unified platform.

The application is developed using the Flask framework, which provides a lightweight and flexible backend for handling user requests, authentication, and booking operations. The system enables users to register, log in, search for travel options, select seats, complete payments, and generate booking confirmations through an intuitive and user-friendly interface.

The deployment of the application on Amazon Web Services (AWS) ensures reliability, scalability, and secure data management. Amazon EC2 is used to host the web application, allowing the system to run continuously and handle multiple users simultaneously. Amazon DynamoDB is utilized for storing user information and booking records, providing fast and efficient data retrieval. Additionally, Amazon SNS is integrated to send real-time notifications to users regarding booking confirmations and cancellations, improving communication and user experience.

Through the use of IAM roles and policies, secure interaction between AWS services is maintained, ensuring that the application follows best practices in cloud security. The integration of Git and GitHub for version control also supports proper project management and collaborative development.

Overall, the TravelGo system successfully demonstrates the practical implementation of a cloud-enabled travel management solution. The project highlights how cloud computing technologies can enhance the efficiency, accessibility, and scalability of modern web applications. By combining user-friendly design with powerful cloud services, TravelGo provides a reliable platform for managing travel reservations while showcasing the potential of cloud-based solutions in real-world applications.